ACTSC 372 Investment Science and Corporate Finance

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0 The Role of Financial Intermediaries

Lecture 1, 2025/09/03

What is financial investment?

- Ownership / loan contracts helped by financial intermediaries.

Role of Financial Intermediaries

- **Size intermediation**: the bank gets small lenders together with the “big” borrowers.
- **Term intermediation**: between short term lenders and long term borrowers.
- **Risk intermediation**: pooling resources reduces individual risk.

Consumer Smoothing: see handwritten notes.

1 Investment Rules

1.1 The NPV Rule

Question: Given a choice of projects, what project should a rational investor choose?

Definition 1.1 (**Project**).

A **project** is a collection of cash flows $\{C_0, C_1, C_2, \ldots\}$ where C_t = cash flow at time t.

Note.

- $C_t < 0 \implies$ cost.
- $C_t > 0 \implies$ revenue.

Definition 1.2 (**Pure Investment Project**).

A **pure investment project** is a project where $C_0 < 0$ and $C_t \geq 0$ for all $t > 0$.

Definition 1.3 (**Pure Financing Project**).

A **pure financing project** is a project where $C_0 > 0$ and $C_t \leq 0$ for all $t > 0$.

Definition 1.4 (**Independent Projects**).

A and B are **independent projects** if the cash flows of A do not affect the cash flows of B.

Note. Between projects A and B, we can choose:

- one,
- both,
- or neither.

Definition 1.5 (**Mutually Exclusive Projects**).

A and B are **mutually exclusive projects** if choosing $A \implies$ we cannot choose B and vice versa (we cannot choose both).

Example. Given $C_0, C_1, C_2, \ldots$. How to decide if this is a "good" project?

Proof. See NPV rule below. $\square$

Definition 1.6 (Net Present Value (NPV)).

The **NPV** of a project is

$$NPV = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \cdots = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

where units = dollars.

Definition 1.7 (NPV Rule).

- For independent projects, choose all projects with $NPV > 0$.
- For mutually exclusive projects, choose the project with the highest NPV, as long as $NPV > 0$.

Example. Let $C_0 = -100, C_1 = 110, C_t = 0$ for $t \geq 2$, and $r = 8\%$. Then,

$$NPV = -100 + \frac{110}{1.08} > 0.$$

So, we should accept this project.

Example. Let $C_0 = 100, C_1 = -110$, and $r = 8\%$. Then,

$$NPV = 100 - \frac{110}{1.08} < 0.$$

So, we should reject this "bad" project.

What is r? $\implies$ r = "opportunity cost" of the dollar.

Advantages of NPV rule:

- NPV takes into account the time value of money.
- NPV takes into account cash flows, not accounting incomes.

Example. Let $C_0 = 100, C_1 = 115, r = 10\%$. The project has positive NPV and the owner will be happy if the project is taken.

Disadvantages of NPV rule:

- The cash flows are uncertain.
- What should be the right "r"? (see later: CAPM)

1.2 Review of ACTSC 231

Some important cash flows to recall from ACTSC 231:

- Perpetuity.
- Perpetuity with growth.
- Annuity.

Definition 1.8 (**Perpetuity, Perpetuity Factor**).

A **perpetuity** is an annuity with no end dates (infinite # of payments). Let $C_t = C$ for all $t \geq 1$. Then,

$$\begin{aligned}
PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots \\
&= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots\right] \\
&= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1}{1+r}} \\
&= \frac{C}{r}.
\end{aligned}$$

The **perpetuity factor** is $\frac{1}{r}$.

Definition 1.9 (Perpetuity with Growth).

Let $C_t = C(1+g)^{t-1}$ for $t \geq 1$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \cdots \\ &= \frac{C}{1+r}\left[1 + \frac{1+g}{1+r} + \left(\frac{1+g}{1+r}\right)^2 + \cdots\right] \\ &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1+g}{1+r}} \\ &= \frac{C}{r-g}, \quad g < r. \end{aligned}$$

Definition 1.10 (Annuity, Annuity Factor).

Let $C_t = C$ for $1 \leq t \leq n$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C}{(1+r)^n} \\ &= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \cdots + \frac{1}{(1+r)^{n-1}}\right] \\ &= \frac{C}{1+r} \cdot \frac{1 - \left(\frac{1}{1+r}\right)^n}{1 - \frac{1}{1+r}} \\ &= C \cdot \frac{1-(1+r)^{-n}}{r}. \end{aligned}$$

The **annuity factor** is $\frac{1}{r}\left[1 - \left(\frac{1}{1+r}\right)^n\right]$.

1.3 Other Investment Rules

Definition 1.11 (Payback Period).

The **payback period** is the # of years required to recoup the initial investment (ignoring the time value of money).

Example. Let $C_0 = -300, C_t = 100$ for $t \geq 1$. The payback period is 3 years.

Note. The units of payback period is years.

Definition 1.12 (**Payback Period Rule**).

Decide on a cutoff period, T^*. If the payback period $\leq T^*$, accept; otherwise, reject it.

Disadvantages of Payback Period Rule:

- It does not take into account the time value of money.
- The cutoff is arbitrary.
- Negative NPV may be accepted.
- Positive NPV may be rejected.

Definition 1.13 (**Discounted Payback Period**).

The **discounted payback period** is the payback period, but without ignoring the time value of money.

Example. Let $C_0 = -300, C_t = 100$ for $t \geq 1$, and $r = 12.5\%$. We are solving $\sum_{t=1}^{T} \frac{C_t}{(1+r)^t} = -C_0$. This implies that the discounted payback period, T, is 4 years.

Definition 1.14 (**Internal Rate of Return (IRR)**).

The **internal rate of return** (**IRR**) of a project is the cost of capital such that the NPV is zero.

Example. Let $C_0 = -100, C_1 = 112, r = 10\%$. We have two ways to determine if we should accept this project:

(1) Calculate NPV: $NPV = -100 + \frac{112}{1.1} > 0 \implies$ accept.

(2) The project has a return of 12%. Since it is bigger than $r = 10\%$, we should accept it.

Question: How to solve for the IRR?

Answer: Calculate the NPV and equal it to zero, then solve for r^*. That is,

$$\sum_{t=0}^{T} \frac{C_t}{(1+r^*)^t} = 0.$$

Definition 1.15 (IRR Rule).

Calculate the iRR of a project.

- For independent projects, choose all projects with IRR $> r$.
- For mutually exclusive projects, choose the project with the highest IRR (as long as IRR $> r$).

Advantages of IRR rule:

- In order to calculate IRR, we do not need r.
- It usually agrees with the NPV rule.

Disadvantages of IRR rule:

The first two disadvantages are for independent projects.

(1) IRR cannot distinguish between investment and financing.

Note. $(-1)\sum \frac{C_t}{(1+r^*)^t} = 0(-1)$, the solution remains the same.

Example. Let $C_0 = -100, C_1 = 112, r = 10\%$. The IRR is 12% and we should accept this project.

Let $C_0 = 100, C_1 = -112, r = 10\%$. The IRR is still 12% but we should reject this project.

(2) Polynomial equation problem: finding the IRR = solving a polynomial equation, i.e.

$$C_0 + \frac{C_1}{1+r^*} + \frac{C_2}{(1+r^*)^2} + \cdots + \frac{C_T}{(1+r^*)^T} = 0.$$

However, we could have no roots, multiple roots, imaginary roots, or complex roots.

Example. Let $C_0 = 0, C_1 = 100, C_t = 0$ for all $t \geq 2$. The IRR $= \infty$.

Example. Let $C_0 = -100, C_1 = 230, C_2 = -132$. Then,

$$-100 + \frac{230}{1+r^*} - \frac{132}{(1+r^*)^2} = 0 \implies r^* = 10\% \text{ or } 20\%.$$

This is problematic, since which IRR should we use?

We will need the following result to deal with the multiple IRR problem.

Theorem 1.1 (**Descartes' Rule of Signs**).

The # of positive roots of a polynomial is $\leq$ # of "sign switches" of the coefficients.

Example. For the previous example where $C_0 = -100, C_1 = 230, C_2 = -132$, there are 2 sign switches. Thus, there are at most 2 positive roots.

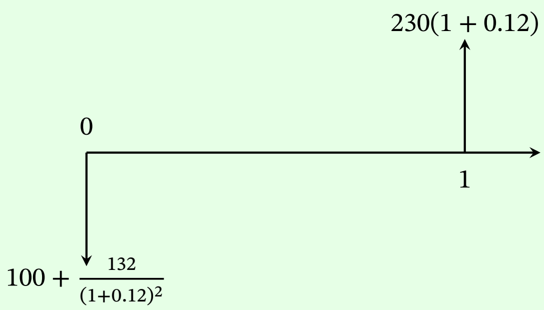
We have the following MIRR to avoid multiple solutions.

Definition 1.16 (**Modified Internal Rate of Return (MIRR)**).

The **modified IRR**

- brings all negative cash flows to time $t = 0$, and
- brings all positive cash flows to time $t = T$,

Example. Let $C_0 = -100, C_1 = 230, C_2 = -132$, and $r = 12\%$. Then,



Calculate the IRR of the modified project (MIRR):

$$-\left(100 + \frac{132}{(1+0.12)^2}\right) + \frac{230(1+0.12)}{1+r^*} = 0 \implies r^* = 12.36\%.$$

Note. We only have 1 "sign switch" in the modified project, so MIRR is unique. Then, we can compare MIRR with r, and choose the project with MIRR $> r$. But we need to know today's interest rate r.

Disadvantages of IRR rule (continued):

For mutually exclusive projects (i.e. we can only choose one project.):

(3) Scaling problem: the IRR is biased towards "smaller" projects.

Example.

Project A: $C_0 = -1, C_1 = 2$.

Project B: $C_0 = -100, C_1 = 150$.

Note that $\text{IRR}_A = 100\%$ and $\text{IRR}_B = 50\%$. However, project B is a better one.

Note. $\sum \frac{kC_t}{(1+r)^t} = 0$. IRR remains the same if you multiply by a constant.

Lecture 3, 2025/09/10

Here are some ways to deal with the scaling problem:

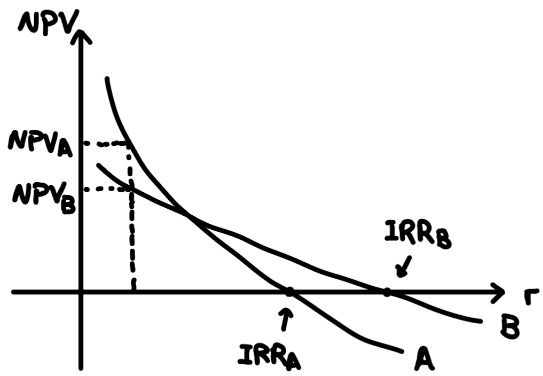
- Calculate the NPV and choose the higher one. In the example, $\text{IRR}_A > \text{IRR}_B$, but $\text{NPV}_A = 1 < \text{NPV}_B$.
- If the IRR picks the "smaller" project, then we do **incremental analysis**.

  Example. Incremental project: $C = B - A$ with $C_0 = -99, C_1 = 148$.

  If the IRR of the incremental project $< r \implies$ stick with the smaller project. If $> r \implies$ go with the bigger project.

(4) Timing problem.

Consider pure investment projects.



According to the IRR rule, B would be chosen, but A is better because $NPV_A > NPV_B$.

Example.

Project short: $C_0 = -300, C_1 = 345, C_t = 0$ for $t \geq 2$.

Project long: $C_0 = -200, C_4 = 330, C_t = 0$ for $t \neq 0, 4$.

Note that

$$IRR_{\text{short}} = 15\%, \quad NPV_{\text{short}} = 13.64$$

$$IRR_{\text{long}} = 13.34\%, \quad NPV_{\text{long}} = 25.39.$$

Note (**Recall from ACTSC 231: YTM of a Bond**).

A bond is a loan in exchange for regular coupon payments and the redemption value at maturity. We have

$$P = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}$$
$$0 = -P + \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C+F}{(1+r)^N}.$$

Note that r is the interest rate that equates NPV to 0. So YTM = IRR of the bond.

Example. Let $C_0 = -100, C_1 = 10, C_2 = 110$ with $r = 10\%$. We can think of this as a bond with $P = 100, C = 10, F = 110, N = 2$. The IRR is 10%.

Also note that if the money was put in a bank, then we would have $C_0 = -100, C_2 = 121$. The extra \$1 is due to the \$10 invested at 10% for another year.

IRR assumes that all cash flows can be reinvested at the IRR itself.

A more realistic assumption that cash flows can be reinvested at r, not IRR.

Definition 1.17 (**Profitability Index (PI)**).

The **profitability index** is

$$PI = \frac{PV \text{ of future cash flows}}{\text{Initial Investment}} = \frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0}.$$

Definition 1.18 (**PI Rule**).

- For independent projects, choose all projects with $PI > 1$.
- For mutually exclusive projects, choose the project with the highest PI with $PI > 1$.

Proposition 1.2. For independent projects, PI agrees with the NPV rule.

Proof. Note that

$$\frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0} > 1 \implies NPV > 0.$$

□

Disadvantages of PI rule:

For mutually exclusive projects:

(1) Scaling problem (same as the IRR rule).

Example. Let $r = 10\%$.

Project A: $C_0 = -10, C_1 = 15, C_2 = 40$.

Project B: $C_0 = -20, C_1 = 70, C_2 = 10$.

Note that

$$PI_A = 4.67, \quad NPV_A = 36.7$$
$$PI_B = 3.60, \quad NPV_B = 51.9.$$

Note that $PI_A > PI_B$, but $NPV_A < NPV_B$. Let's consider the incremental project $C = B - A$ with $C_0 = -10, C_1 = 55, C_2 = -30$.

If the incremental project has $PI > 1$, then choose the bigger project; otherwise, choose the smaller project.

Summary of Investment Rules: NPV rule is the best.

1.4 Equivalent Annual Costs (EAC)

Example (Machines with Unequal Lives).

Let $r = 10\%$ and consider the following two machines.

A: \$500 to buy, \$120/year to operate, lasts for 3 years.

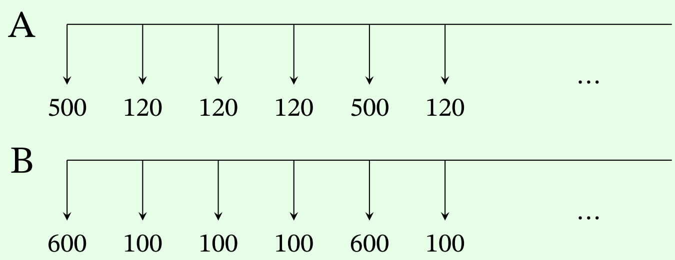
B: \$600 to buy, \$100/year to operate, lasts for 4 years.

Note that we have to use the machine (so we buy another one after it breaks down).

We can calculate the NPVs to get

$$NPV_A = -798.42, \quad NPV_B = -916.94.$$

However, the actual cash flows are different. We have the following cash flows:



Find the EAC of each machine.

Proof. This is equivalent to solving for x in $NPV_A = x \cdot a_{\overline{3}|10\%}$ AND $NPV_B = x \cdot a_{\overline{4}|10\%}$. Then, we get

$$EAC_A = \frac{NPV_A}{a_{\overline{3}|10\%}}, \quad EAC_B = \frac{NPV_B}{a_{\overline{4}|10\%}}.$$

We choose the one with lower EAC! $\square$

Goal: Calculate NPV, cash flows, r, in the right way!

2 Capital Budgeting

2.1 Rules for Calculating NPVs

Remember the folllowing rules when calculating NPVs:

(1) Ignore sunk costs, i.e. costs we have to incur with/without the project.

> **Example.** Thaler Pizza experiment.

(2) Include opportunity costs, i.e. cost of the next best alternative.

Lecture 4, 2025/09/15

(3) Be aware of externalities, i.e. effects on other projects. One project can affect cash flows from other projects.

> **Example.**
> - Positive externality: synergy.
> - Negative externality: erosion.

(4) Be aware of the difference between "real" and "nominal".

Note.

- Nominal Interest (i): in dollars.
- Real interest rate (r): amount of extra goods you can buy.
- Rate of inflation (π): increase with prices.

> *Example.* Suppose you have \$1 today and price of the good is \$1. If you put \$1 in the bank, it will become $1 + i$ dollars in one year. Price of the good in one year will be $1 + \pi$. So the # of goods you can buy next year is
>
> $$\frac{1 + i}{1 + \pi} = 1 + r.$$

The # of extra units of the good you can buy is (Fisher's Law):

$$r = \frac{1+i}{1+\pi} - 1 = \frac{i-\pi}{1+\pi}.$$

We can also approximate r by

$$r \approx i - \pi.$$

Rule for Calculating Cash Flows

- Be consistent. If we are using real cash flows, use real interest rates, otherwise use nominal cash flows and nominal interest rates.

  Note. Nominal cash flows to real cash flows:

  $$\text{Divide by } (1+\pi)^t.$$

  Real cash flows to nominal cash flows:

  $$\text{Multiply by } (1+\pi)^t.$$

Example. Suppose all cash flows are nominal. Let $C_0 = -1000, C_1 = 600, C_2 = 650, i = 14\%, \pi = 5\%$. Calculate the NPV.

Proof.
Method 1: Use nominal.

$$NPV = -1000 + \frac{600}{1+0.14} + \frac{650}{(1+0.14)^2}.$$

Method 2: Find the real interest rate from Fisher's Law:

$$1 + r = \frac{1+i}{1+\pi}.$$

Let C_0' be the real cash flow at time 0, then

$$\begin{aligned} C_0' &= C_0 \\ C_1' &= \frac{C_1}{1+\pi} \\ C_2' &= \frac{C_2}{(1+\pi)^2}. \end{aligned}$$

Then,

$$NPV = C_0' + \frac{C_1'}{1+r} + \frac{C_2'}{(1+r)^2}.$$

We should get the same answer using both methods. □

Remark (**Why does this work?**).

Suppose all cash flows are nominal. Then,

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+i} + \frac{C_2}{(1+i)^2} + \cdots \\ &= C_0 + \frac{\frac{C_1}{1+\pi}}{\frac{1+i}{1+\pi}} + \frac{\frac{C_2}{(1+\pi)^2}}{\left(\frac{1+i}{1+\pi}\right)^2} + \cdots \\ &= C_0' + \frac{C_1'}{1+r} + \frac{C_2'}{(1+r)^2} + \cdots. \end{aligned}$$

(5) Be aware of depreciation for cash flows from accounting statements.

Example.

Revenue	**Cost**	**Depreciation**	**Taxes**
1500	700	600	34%

Depreciation is a tax-deductible expense! For the taxable income,

$$\begin{aligned} \text{Taxable Income} &= \text{Revenue} - \text{Cost} - \text{Depreciation} \\ &= 1500 - 700 - 600 = 200. \end{aligned}$$

Also,

$$\text{Tax paid} = 0.34 \times 200 = 68.$$

Then,

$$\text{Cash Flow} = \text{Revenue} - \text{Cost} - \text{Tax paid} = 1500 - 700 - 68 = 732.$$

General approach: Let the cash flow be CF_t. Then,

$$\begin{aligned} CF_t &= \text{Revenues} - \text{Costs} - \text{Taxes} \\ &= (R_t - \text{Cost}_t) - (R_t - \text{Cost}_t - D)T_c \\ &= (R_t - \text{Cost}_t)(1 - T_c) + DT_c. \end{aligned}$$

The term $(R_t - \text{Cost}_t)(1 - T_c)$ is called the **after tax operating cash flow (ATOCF)**. Note that the term DT_c is the tax break we get because we can substract depreciation. And DT_c is the **capical cost allowance tax shield (CCATS)**.

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \cdots \frac{C_t}{(1+r)^t} \\ &= C_0 + \frac{C_1}{1+r} + \cdots + \frac{(\text{ATOCF} + \text{CCATS})}{(1+r)^t} \\ &= C_0 + PV\text{ATOCF} + PV\text{CCATS}. \end{aligned}$$

Note that $PV(\text{ATOCF})$ is the PV without subtracting depreciation, and $PV(\text{CCATS})$ is the PV of all tax breaks.

Note. To see the above, we have

$$\begin{aligned} NPV &= C_0 + \frac{C_1}{1+r} + \cdots + \frac{C_t}{(1+r)^t} + \cdots \\ &= C_0 + \frac{(\text{ATOCF})_1 + (\text{CCATS})_1}{1+r} + \cdots + \frac{(\text{ATOCF})_t + (\text{CCATS})_t}{(1+r)^t} + \cdots \\ &= C_0 + \sum_{t=1}^{\infty} \frac{(\text{ATOCF})_t}{(1+r)^t} + \sum_{t=1}^{\infty} \frac{(\text{CCATS})_t}{(1+r)^t}. \end{aligned}$$

Question: How is depreciation (D) calculated?

Example. Suppose $d = 20\%$. A machine costs \$1 million dollars.

Year	UCC Beginning	Depreciation (D)	UCC End
1	\$1,000,000	\$200,000	\$800,000
2	\$800,000	\$160,000	\$640,000
3	\$640,000	\$128,000	\$512,000

Note that $D = d \cdot (\text{UCC Beginning})$, where UCC is the **undepreciated capital cost**.

Half-Year Rule: if you buy an asset, only 50% of the purchase price can be added to the first year. The remaining 50% can be added in the second year.

Example (Half-Year Rule).
Suppose $d = 20\%$ and a machine costs \$1 million dollars.

Year	UCC Beginning	Depreciation (D)	UCC End
1	\$500,000	\$100,000	\$400,000
2	\$900,000	\$180,000	\$720,000
3	\$720,000	\$144,000	\$576,000

Theorem 2.1 (PVCCATS Formula).
We have

$$PV(\text{CCATS}) = \frac{C \cdot d \cdot T_c}{r + d} \cdot \frac{1 + 0.5r}{1 + r} - \frac{S \cdot d \cdot T_c}{r + d} \cdot \frac{1}{(1 + r)^n}$$

where

- C = cost of the asset.
- d = rate of depreciation.
- n = lifetime of the asset.
- S = salvage value of the asset.

2.2 Capital Budgeting Under Uncertainty

Example. MATHSOC is thinking of installing a solar panel:

- Cost: \$1 million.
- Annual operating cost: $10K$.
- Lifetime: 15 years.
- Salvage: $100K$.
- Revenues: \$100/hour.
- # of sunny hours: 1500.
- $r = 10\%$.
- No taxes.

Find the NPV.

Proof. We have

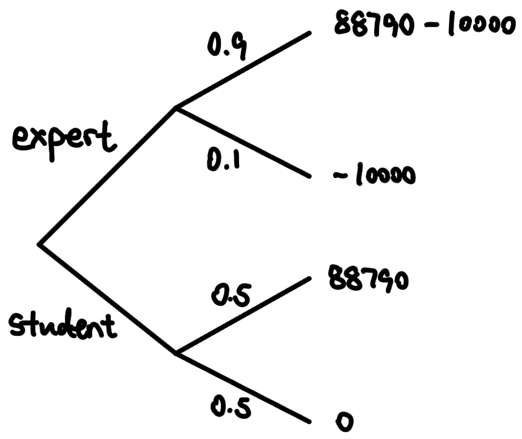
$$NPV = -1000000 - 10000a_{\overline{15}|10\%} + 150000a_{\overline{15}|10\%} + \frac{100000}{(1.1)^{15}} = 88790.$$

□

Decision Tree

For the MATHSOC example, installation may not work. Consider two options:

- Hire an expert: cost \$10000, success rate 90%.
- Hire a student: cost \$0, success rate 50%.



Lecture 5, 2025/09/17

There are two types of nodes in a decision tree:

- Probability nodes: calculate expected values.
- Decision nodes: choose the "better" path.

Example (**Continued**).

At the expert node, the expected value is $0.9(88790) + 0.1(-10000) = 76911$.

At the student node, the expected value is $0.5(88790) + 0.5(-10000) = 39395$.

Since $76911 > 39395$, we should hire the expert.

Scenario Analysis

For the MATHSOC example, what if the # of sunny hours $\neq 1500$? Suppose # of sunny hours are $1100, 1500, 1900$ with probability $\frac{1}{3}$ each.

We calculate the expected NPV, i.e. $\mathbb{E}[NPV] = \frac{1}{3}(NPV_{1100} + NPV_{1500} + NPV_{1900})$.

Monte Carlo Simulation

Suppose sunny hours $\sim \mathrm{N}(1500, 200)$. We simulate n observations from this distribution, and calculate NPVs for each outcome, and looke at the distribution of NPVs.

Sensitivity Analysis

How sensitive is our NPV to different variables? Let $S = \#$ of sunny hours. Then, we want to find $\frac{\mathrm{d}NPV}{\mathrm{d}S} \implies$ how accurate we need to be with our estimates.

Break-Even Analysis

Break-even point is the value of the variable that makes NPV $= 0$.

Definition 2.1 (**Contribution Margin**).

The **contribution margin** is the increase in net income per unit increase in sales, i.e.

$$\text{Contribution Margin} = (\text{Unit Revenue} - \text{Unit Cost}) \cdot (1 - T_c)$$

where T_c is the corporate tax rate.

Example. The Tim Hortons at SLC sells donuts. Assume no taxes.

- Donuts: \$1.20.
- Cost of donuts: \$0.70.
- Fixed cost: \$100,000.

Contribution Margin $= 1.20 - 0.70 = 0.50$. Then, the break-even point for # of donuts sold is $\frac{100000}{0.50} = 200000$, i.e. we need to sell 200000 donuts to cover the fixed cost.

Example. Same setting as above example, but with tax = 40%. Then,

$$\text{Contribution Margin} = (1.20 - 0.70)(1 - 0.4) = 0.30.$$

Then, the break-even point for # of donuts sold is $\frac{100000}{0.30} \approx 333333$.

Example. The IRR is the break-even interest rate!

Example. The discounted payback period is the break-even time!

For the next few lectures, we will focus on the following question.

Question: What is the right "r" for a project? (CAPM)

2.3 Utility Theory

Before we get there, let's consider the following question.

Question: How do rational human beings evaluate uncertain outcomes?

Definition 2.2 (Lottery).

A lottery X is a random variable whose outcomes are dollars.

Example. Consider the lotteries X and Y with the following PMFs:

X	100	25
$\mathbb{P}(X = x)$	0.5	0.5

Y	81	49
$\mathbb{P}(Y = y)$	0.5	0.5

One way to compare lotteries is to compare their expected values.

Example (**ST. Petersburg Paradox**).

Let X be the amount of money you win from the following lottery. A fair coin is tossed until it comes up heads. If the first head appears on the n-th toss, you win 2^n dollars.

X	Probability
2	$\frac{1}{2}$
4	$\frac{1}{4}$
8	$\frac{1}{8}$
16	$\frac{1}{16}$
$\vdots$	$\vdots$

The expected value of the lottery is

$$\mathbb{E}[X] = \sum x_i p_i = \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \cdots = 1 + 1 + 1 + \cdots = \infty.$$

So you should be willing to pay any finite amount to play this game. However, in reality, most people are not willing to pay a lot of money to play this game.

John-von-Neumann: Expected Utility Theory

We use the preference relation $\succeq$ to denote "is at least as good as". That is,

$$x \succeq y \implies x \text{ is at least as good as } y.$$

Question: What assumptions should $\succeq$ follow for rational agents?

Assumptions

(1) Completeness: $\forall X, Y$, either $X \succeq Y$ or $Y \succeq X$ (or both).

(2) Transitivity: $\forall X, Y, Z$, if $X \succeq Y$ and $Y \succeq Z$, then $X \succeq Z$.

(3) Independence: $\forall X, Y, Z$ and $\forall p \in (0,1)$, $X \succeq Y \iff pX + (1-p)Z \succeq pY + (1-p)Z$.

Theorem 2.2. If $\succeq$ satisfies the above three assumptions, then $\exists$ a function $u(\cdot)$ such that $\forall X, Y$,

$$X \succeq Y \iff \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)].$$

Note. $u(\cdot)$ is called the **von-Neumann Morgenstern utility function**. It is the utility you get after consuming x dollars.

Example. Consider the following lotteries with $u(x) = \sqrt{x}$.

X	100	25
$\mathbb{P}(X = x)$	0.5	0.5

Y	81	49
$\mathbb{P}(Y = y)$	0.5	0.5

For lottery X,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{25} = 5 + 2.5 = 7.5.$$

For lottery Y,

$$\mathbb{E}[u(Y)] = 0.5\sqrt{81} + 0.5\sqrt{49} = 4.5 + 3.5 = 8.$$

2.4 Risk Aversion

Definition 2.3 (**Risk Averse**).

A person is a **risk averse** if (for any lottery), they prefer the expected values for sure over the lottery itself.

Definition 2.4 (**Risk Neutral**).

A person is **risk neutral** if they only care about the expected values.

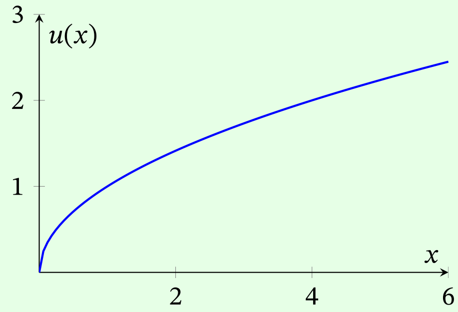
Definition 2.5 (**Risk Loving**).

A person is **risk loving** if they prefer the lottery over the expected values.

Proposition 2.3. If the von-Neumann Morgenstern utility function $u(\cdot)$

- is concave, then the person is risk averse.
- is linear, then the person is risk neutral.
- is convex, then the person is risk loving.

Example. Let $u(x) = \sqrt{x}$. Note that $u''(x) < 0$, so the person is risk averse.



Decreasing marginal utility over money $\implies$ Risk averse, because the more money you have, the less utility you get from an extra dollar.

Say we have lotteries X and Y with the following PMFs:

X	100	0
$\mathbb{P}(X = x)$	0.5	0.5

Y	50
$\mathbb{P}(Y = y)$	1

Then,

$$\mathbb{E}[u(X)] = 0.5\sqrt{100} + 0.5\sqrt{0} = 5$$
$$\mathbb{E}[u(Y)] = 1 \cdot \sqrt{50} \approx 7.07.$$

If $u(x) = x$, then $\mathbb{E}[u(X)] = 0.5 \cdot 100 + 0.5 \cdot 0 = 50$ and $\mathbb{E}[u(Y)] = 1 \cdot 50 = 50$. So the person is risk neutral.

If $u(x) = x^2$, then $\mathbb{E}[u(X)] = 0.5 \cdot 100^2 + 0.5 \cdot 0^2 = 5000$ and $\mathbb{E}[u(Y)] = 1 \cdot 50^2 = 2500$. So the person is risk loving.

Lecture 6, 2025/09/22

Recall that a lottery X is a random variable with dollars as outcomes, where x = outcome of X. The PMF of X is $f(x) = \mathbb{P}(X = x)$.

Let $U(X)$ be the utility you get from lottery X. Recall Theorem 2.2:

If a human being is rational, $\exists u(\cdot)$ over money such that

$$X \succeq Y \iff \mathbb{E}[u(X)] \geq \mathbb{E}[u(Y)]$$

where $U(X)$ = utility one gets from lottery $X = \mathbb{E}[u(X)]$.

Proposition 2.4.

(1) A person is risk averse if $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$ for all X.

(2) A person is risk neutral if $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$ for all X.

(3) A person is risk loving if $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$ for all X.

Definition 2.6 (Certainty Equivalent).

The **certainty equivalent** of a lottery X is $CE(X)$ = dollar value (for sure) that makes you equally happy as the lottery, i.e.

$$u(CE(X)) = \mathbb{E}[u(X)].$$

Example. Let X be a lottery with the following PMF:

X	125	64	0
$\mathbb{P}(X = x)$	1/3	1/3	1/3

Let $u(x) = x^{\frac{1}{3}}$. Then,

$$\mathbb{E}[u(X)] = \frac{1}{3}125^{\frac{1}{3}} + \frac{1}{3}64^{\frac{1}{3}} + \frac{1}{3}0^{\frac{1}{3}} = 3.$$

Then, the certainty equivalent is a solution to $u(x) = 3$, i.e.

$$x^{\frac{1}{3}} = 3 \implies x = 27.$$

Proposition 2.5.

(1) Risk averse: $CE(X) < \mathbb{E}[X]$.

(2) Risk neutral: $CE(X) = \mathbb{E}[X]$.

(3) Risk loving: $CE(X) > \mathbb{E}[X]$.

Degree of Risk Aversion

Note that u'' measures how risk averse a person is. We use the following measures.

Definition 2.7 (**Absolute Risk Aversion (ARA)**).

The **absolute risk aversion (ARA)** is

$$\text{ARA} = -\frac{u''(x)}{u'(x)}.$$

This is also called the Arrow-Pratt measure of absolute risk aversion.

Definition 2.8 (**Relative Coefficient of Risk Aversion (RRA)**).

The **relative coefficient of risk aversion (RRA)** is

$$\text{RRA} = -\frac{xu''(x)}{u'(x)} = x \cdot \text{ARA}\,.$$

Example. Let $u(x) = -e^{-\alpha x}$. Calculate the ARA.

Proof. Note that

$$u'(x) = \alpha e^{-\alpha x}, \qquad u''(x) = -\alpha^2 e^{-\alpha x}.$$

Then,

$$\text{ARA} = -\frac{u''(x)}{u'(x)} = -\frac{-\alpha^2 e^{-\alpha x}}{\alpha e^{-\alpha x}} = \alpha.$$

□

Example. Let $u(x) = x^\alpha$ for $0 < \alpha < 1$. Calculate the RRA.

Proof. Note that

$$u'(x) = \alpha x^{\alpha-1}, \qquad u''(x) = \alpha(\alpha-1)x^{\alpha-2}.$$

Then,

$$\text{RRA} = -\frac{xu''(x)}{u'(x)} = -\frac{x\alpha(\alpha-1)x^{\alpha-2}}{\alpha x^{\alpha-1}} = 1-\alpha.$$

□

2.5 Mean-Variance Utility

This is a different approach to measure risk aversion.

Definition 2.9 (**Mean-Variance Utility**).
Let X be a lottery with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathrm{Var}(X)$. The **mean-variance utility** is

$$U(X) = \mu - \frac{A}{2}\sigma^2$$

where A is a known constant.

Note. All mean-variance utility are risk averse.

In fact, A is the **coefficient of risk aversion**. The higher A is, the more risk averse the person is.

Remark (**Summary of Two Approaches**).

(1) $U(X) = \mathbb{E}[u(X)]$.

(2) $U(X) = \mu - \frac{A}{2}\sigma^2$.

Question: Are these two approaches related?

Suppose we have a von-Neumann Morgenstern utility function

$$u(x) = x - \alpha x^2.$$

Then, $u'(x) = 1 - 2\alpha x$. Note that we want $u'(x) > 0 \implies x < \frac{1}{2\alpha}$, i.e. we are looking at "small" gambles. Note that a von-Neumann Morgenstern agent maximizes

$$\mathbb{E}[u(X)] = \mathbb{E}\left[X - \alpha X^2\right] = \mathbb{E}[X] - \alpha\mathbb{E}\left[X^2\right].$$

Note that

$$\mathrm{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2 \implies \mathbb{E}\left[X^2\right] = \mathrm{Var}(X) + (\mathbb{E}[X])^2.$$

Then,

$$\mathbb{E}[u(X)] = \mu - \alpha(\sigma^2 + \mu^2) = \mu - \alpha\sigma^2 - \underbrace{\alpha\mu^2}_{\approx 0} \approx \mu - \alpha\sigma^2.$$

Example. Suppose the von-Neumann Morgenstern utility function is

$$u(x) = -e^{-\alpha x}.$$

Let $X \sim \mathrm{N}(\mu, \sigma^2)$. Then, this person is also a mean-variance utility maximizer.

Proof. We will do the proof in the next tutorial. □

We need to find μ and σ^2 of a lottery. We estimate μ by the following:

$$\hat{\mu} = \frac{1}{n} \sum_{t=1}^{n} r_t,$$

where r_t is the return at time t.

3 Capital Asset Pricing Model (CAPM)

3.1 Risk and Return

Question: How are the r_t's calculated?

Let P_t be the price of an asset today. Then, P_{t+1} is the price of the asset in 1 year. Let D_t be the dividends you get this year. Then, the return is

$$r_t = \frac{(P_{t+1} + D_t) - P_t}{P_t} = \frac{P_{t+1} - P_t}{P_t} + \frac{D_t}{P_t}$$

where $\frac{P_{t+1}-P_t}{P_t}$ is the **price yield** and $\frac{D_t}{P_t}$ is the **dividend yield**.

Example (Holding Period Return).

Suppose the bank offers the following deal:

- 4% for year 1.
- 8% for year 2.
- 12% for year 3.

What is the average return?

Proof. Let the average be x. Then,

$$(1+x)^3 = (1+0.04)(1+0.08)(1+0.12) \implies x = \sqrt[3]{1.04 \times 1.08 \times 1.12} - 1.$$

Note that x is the geometric mean in this case (not arithmetic mean). □

Measure of Volatility

Definition 3.1 (Variance and Standard Deviation of Returns).

The variance of the returns is defined as

$$\sigma^2 = \frac{1}{n-1} \sum (r_t - \mu)^2,$$

Also, $\sigma = \sqrt{\sigma^2}$ is the standard deviation.

Other Measures

Let X and Y be the following two distributions of returns where X has a long left tail and Y has a long right tail. Here, Y is preferred by most investors even though both have the same variance.

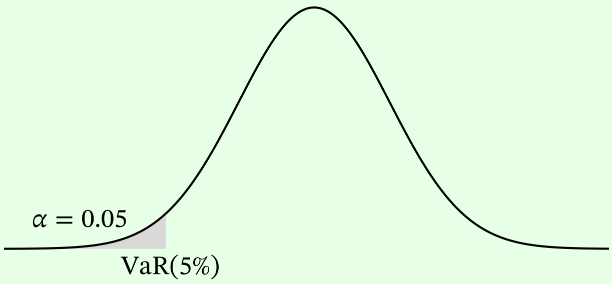
We need to come up with other measures of risk which capture the asymmetry.

Definition 3.2 (**Value At Risk (VaR)**).
$\mathrm{VaR}(\alpha)$, where α is given as that value of x such that

$$\mathbb{P}(X \le x) = \alpha.$$

In other words, VaR is the best case for the worst case scenarios.

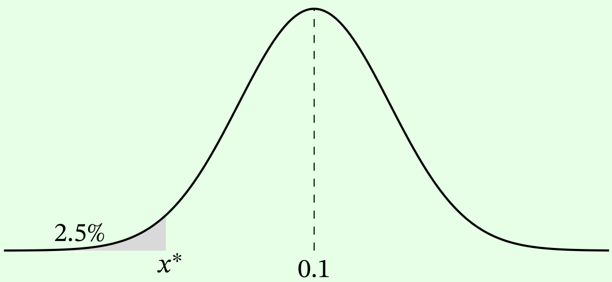
Example. VaR(5%) can be interpreted in the following plot.



Example. Let $X \sim \mathrm{N}(0.1, 0.08^2)$. Find VaR(2.5%).

Proof. Let x^* be such that $\mathbb{P}(X \le x^*) = 2.5\%$. Then,

$$\mathbb{P}\left(\frac{X-\mu}{\sigma} \le \frac{x^*-\mu}{\sigma}\right) = 2.5\% \implies \mathbb{P}(Z \le y^*) = 0.025.$$



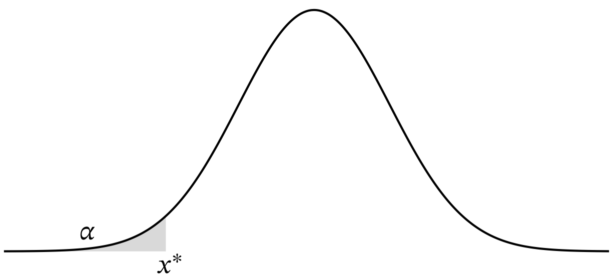
Thus, $y^* = \frac{x^* - \mu}{\sigma} = -1.96 \implies x^* = \mu + \sigma y^* = 0.1 + 0.08 \times (-1.96) = -0.0568$. Hence, $\text{VaR}(2.5\%) = -5.68\%$. $\square$

Definition 3.3 (**Expected Loss (EL)**).

The **expected loss (EL)** or **conditional expected loss (CEL)** is defined as

$$\text{EL}(\alpha) = \text{average return you make, conditional on the fact that } X \leq \text{VaR}(\alpha).$$

Look at the following plot.



Here, $x^* = \text{VaR}(\alpha)$ with $\mathbb{P}(X \leq x^*) = \alpha$. The CEL is the average of the shaded area. i.e.

$$\text{CEL}(\alpha) = \mathbb{E}\left[X \mid X \leq x^*\right].$$

Lecture 7, 2025/09/24

Recall some other measures of volatility:

- VaR.
- EL.
- Downside semi-variance.

Example. Suppose we have the following returns and probabilities.

X (%)	-15	-10	0	10	20
$f(x)$	0.02	0.03	0.5	0.4	0.05

Note that Var(5%) $= -10\%$. For EL(5%),

X (%)	-15	-10
Conditional probs	0.4	0.6

Then, EL(5%) $= -15\% \times 0.4 + (-10\%) \times 0.6 = -12\%$.

Definition 3.4 (**Downside Semi-Variance**).

The **downside semi-variance** is defined as

$$\text{DSV} = \frac{1}{n-1} \sum_{r_i < \mu} (r_i - \mu)^2.$$

3.2 Capital Asset Pricing Model

Question: What portfolio will a risk-averse individual choose?

General Problem: Given n assets, a portfolio is $\mathbf{w} = (w_1, \ldots, w_n)$ where w_i is the proportion of wealth invested in asset i. Note that $\sum_{i=1}^{n} w_i = 1$. What should the w_i's be?

Capital Allocation Line

Suppose we are in a two-asset universe.

- Asset F: risk-free asset with return r_f.
- Asset P: risky asset with μ_p = average return, σ_p = volatility.

Example. Let $r_f = 7\%$, $\mu_p = 15\%$, $\sigma_p = 20\%$. A portfolio here is $(1-y, y)$, where y is the proportion invested in the risky asset and $1-y$ is the proportion invested in the risk-free asset. Suppose we invest \$10000. If $y = 0.3$, then we invest \$7000 in F and \$3000 in P.

The utility function is

$$U(x) = \mu - \frac{A}{2}\sigma^2.$$

Question: What should y be?

Example. Let $r_f = 7\%$, $\mu_p = 15\%$, $\sigma_p = 20\%$. Client 1 wants to invest with an average return of 10%. What should be y?

Proof. Note that

$$(1-y)r_f + y\mu_p = 10\% \implies y = \frac{10\% - 7\%}{15\% - 7\%} = 0.375.$$

□

Client 2 wants the volatility to be $< 15\%$. What should be y?

Proof. Note that the portfolio of Client 2 is

$$\underbrace{C}_{Y} = \underbrace{(1-y)F}_{a} + \underbrace{y}_{b}\underbrace{P}_{X}.$$

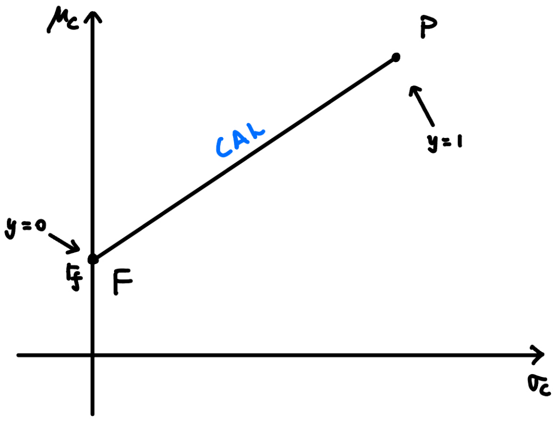
Remark. Recall from STAT 230, let $Y = a + bX$, then $\sigma_Y = |b|\sigma_X$.

Also, F is a constant and P is a random variable. Thus,

$$\sigma_C = y\sigma_P < 15\% \implies y < \frac{15\%}{20\%} = 0.75.$$

□

Question: What are the possible (μ, σ) pairs we can create?



Note that y = proportion invested in the risky asset P.

Question: What happens if y changes?

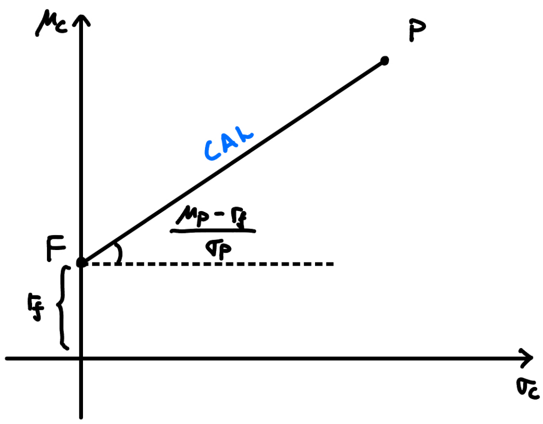
Proof. Note That

$$C = (1-y)F + yP$$
$$\mu_C = \mathbb{E}\left[(1-y)r_f + yP\right] = (1-y)r_f + y\mu_p = r_f + y\underbrace{(\mu_p - r_f)}_{\text{risk premium}}.$$

Also, $\sigma_C = y\sigma_p$. Then,

$$y = \frac{\sigma_C}{\sigma_P} \implies \mu_C = r_f + \frac{\sigma_C}{\sigma_P}(\mu_P - r_f) = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C.$$

□



The slope of the CAL is $\frac{\mu_P - r_f}{\sigma_P}$, which is called the **Sharpe ratio** (or **reward-risk ratio**).

Definition 3.5 (Sharpe Ratio).

The ratio

$$\frac{\mu_P - r_f}{\sigma_P}$$

is called the **Sharpe ratio** or **reward-risk ratio** of the risky asset P.

Investor's Problem

We have the following optimization problem:

$$\max_C \mu_C - \frac{A}{2}\sigma_C^2 \text{ s.t. } C \text{ is on the CAL.}$$

To solve this,

$$\mu_C = r_f + y(\mu_P - r_f), \quad \sigma_C = y\sigma_P.$$

Then, the problem becomes

$$\max_y \left\{ r_f + y(\mu_P - r_f) \right\} - \frac{A}{2}(y\sigma_P)^2.$$

Taking derivative and setting it to 0,

$$\mu_P - r_f - \frac{2Ay\sigma_P^2}{2} = 0 \implies y^* = \frac{\mu_P - r_f}{A\sigma_P^2}.$$

Note. This is why we have $\frac{1}{2}$ in the utility function $U(x) = \mu - \frac{A}{2}\sigma^2$.

Example. Let $f_f = 7\%$, $\mu_P = 15\%$, $\sigma_P = 20\%$, $A = 4$. What is y^*?

Proof. We have

$$y^* = \frac{\mu_P - r_f}{A\sigma_P^2} = \frac{15\% - 7\%}{4 \times (20\%)^2} = 0.5.$$

□

We saw the algebraic way of looking at this portfolio problem. Now, we look at it geometrically.

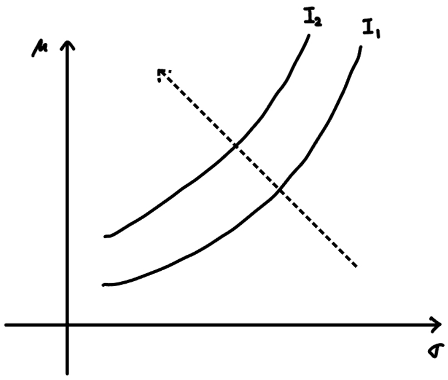
Geometric Approach

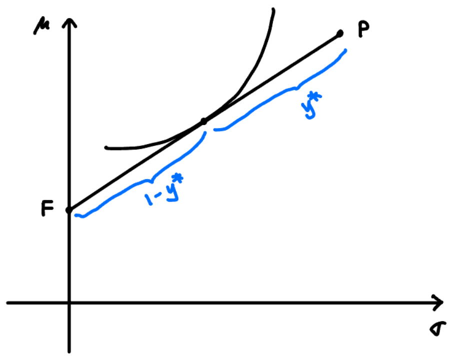
Definition 3.6 (**Indifference Curve**).
An **indifference curve** is defined as

$$\{(\sigma, \mu) : \text{ utility is the same for all points on the graph}\}.$$

That is, all points make you equally happy.

Note. Indifference curves have positive slopes.





Different indifference curves correspond to different levels of utility, and we want to move as top-left as possible. So maximizing happiness is equivalent to finding the tangent to the CAL (the most top-left indifference curve that touches the CAL).

Question: Can the indifference curves ever intersect?

Answer: No, because $A \sim B$ and $B \sim C$ implies $A \sim C$. That is, if A is as happy as B and B is as happy as C, then A is as happy as C.

Remark. $X \sim Y$ means X is indifferent to Y (i.e. they provide the same utility).

Question: Is it possible that $y^* > 1$?

Answer: Yes, we can borrow money and invest more in the risky asset.

Lecture 8, 2025/09/29

Question: For a risk-averse, mean-variance utility agent, what portfolio will they choose?

Case 1: One risk-free asset F and one risky asset P.

We have a portfolio $(1-y, y)$ where y = proportion in the risky asset. For a porfolio C, we have

$$U = \mu_C - \frac{A}{2}\sigma_C^2.$$

Example. Let $r_f = 7\%$, $\mu_P = 15\%$, $\sigma_P = 20\%$. Suppose $y = 0.75$. What are μ_C and σ_C?

Proof. Note that

$$\mu_C = 0.25 \cdot 7\% + 0.75 \cdot 15\% = 13\%, \quad \sigma_C = 0.75 \cdot 20\% = 15\%.$$ □

The feasible set here is a straight line called the **Capital Allocation Line (CAL)**. The equation of the CAL is

$$\mu_C = r_f + \left(\frac{\mu_P - r_f}{\sigma_P}\right)\sigma_C$$

where the slope is the **Sharpe Ratio**.

We had the optimization problem:

$$\max_C \mu_C - \frac{A}{2}\sigma_C^2 \text{ s.t. } C \text{ is on the CAL} \implies y^* = \frac{\mu_P - r_f}{A\sigma_P^2}.$$

Note that y^* depends on the risk premium $\mu_P - r_f$, the volatility σ_P, and the risk aversion A.

Case 2: The two risky asset model.

Two Assets	Mean	SD
A	μ_A	σ_A
B	μ_B	σ_B

where $\sigma_A, \sigma_B > 0$. Let (w_A, w_B) be a portfolio, where w_A is the proportion invested in asset A and w_B is the proportion invested in asset B.

Question: What w_A, w_B should we choose?

For the feasible set, it is no longer linear (in general). We have

$$C = w_A A + w_B B, \quad w_A + w_B = 1.$$

Then,

$$\mu_C = w_A\mu_A + w_B\mu_B, \quad \sigma_C^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \operatorname{Cov}(A, B).$$

Remark (**STAT 230 Review**).
Recall that

$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y.$$

Let $X = \#$ of beers drunk, $Y =$ ACTSC 372 grades. If $x > \mu_X$, it is more likely that $y < \mu_Y$. Thus, $\operatorname{Cov}(X, Y) < 0$. The idea is that $\operatorname{Cov}(X, Y)$ is the average of the product of the deviations from the mean. It is also the average of the product minus the product of the averages.

Proposition 3.1 (**Properties of Covariance**).

(1) $\operatorname{Cov}(X, k) = 0$ for any constant k.

(2) $\operatorname{Cov}(X, X) = \operatorname{Var}(X)$.

(3) Bilinearity: $\operatorname{Cov}(aX + bY, Z) = a\operatorname{Cov}(X, Z) + b\operatorname{Cov}(Y, Z)$ for any constants a, b.

(4) If $X \perp\!\!\!\perp Y$, then $\operatorname{Cov}(X, Y) = 0$. (The converse is not true in general.)

Example (**Counterexample for Converse of (4)**). Let X be such that

X	-1	1
$f(x)$	$\frac{1}{2}$	$\frac{1}{2}$

Let $Y = X^2 = 1$ so that X and Y are not independent. Then,

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X\mu_Y = 0 - 0 \cdot 1 = 0.$$

Note that Cov(X, Y) has units. We define the following to get a unitless measure.

Definition 3.7 (Correlation Coefficient).

The **correlation coefficient** between two random variables X and Y is

$$\rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Note that ρ_{XY} is unit independent.

Proposition 3.2 (Properties of Correlation Coefficient).

(1) $-1 \leq \rho_{XY} \leq 1$ for all X, Y.

(2) If $X \perp\!\!\!\perp Y$, then $\rho_{XY} = 0$.

(3) If $\rho = 0$, then there is no linear relationship between X and Y.

Recall that for any porfolio $C = (w_A, w_B)$, we have

$$\begin{aligned} \mu_C &= w_A \mu_A + w_B \mu_B \\ \sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \operatorname{Cov}(A, B) \\ &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho \sigma_A \sigma_B. \end{aligned}$$

Special Case: $\rho = 1$

Note that

$$\begin{aligned} \mu_C &= w_A \mu_A + w_B \mu_B \\ \sigma_C^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \sigma_A \sigma_B = (w_A \sigma_A + w_B \sigma_B)^2. \end{aligned}$$

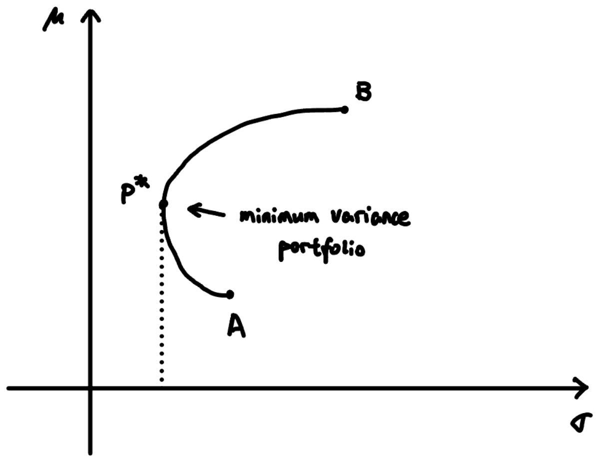
Then, $\sigma_C = w_A \sigma_A + w_B \sigma_B$. Note that σ_C is linear in w_A, w_B. Thus, the feasible set is a straight line.

If $\rho \neq 1$, then $\sigma_C < w_A \sigma_A + w_B \sigma_B$. The feasible set will be bowed towards the top left.

Value of Diversification

Diversification means to invest in multiple assets. This lowers σ for the same μ.

There are limits to diversification! Look at the plot below, p^* = minimum variance portfolio.



Special Case: $\rho = -1$

Note that

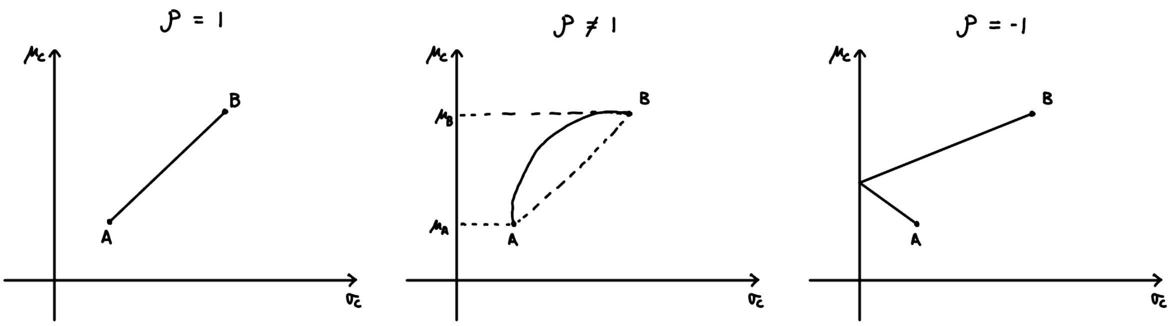
$$\begin{aligned}
\sigma_C^2 &= w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B(-1)\sigma_A\sigma_B \\
&= (w_A\sigma_A - w_B\sigma_B)^2 \\
\implies \sigma_C &= |w_A\sigma_A - w_B\sigma_B|.
\end{aligned}$$

Choosing

$$w_A = \frac{\sigma_B}{\sigma_A + \sigma_B}, \quad w_B = \frac{\sigma_A}{\sigma_A + \sigma_B}$$

gives $\sigma_C = 0$. All the risk can be diversified away (i.e. $\sigma_C = 0$). In general, it is <u>not true</u>.

Here is a summary of different ρ values:



Note. For $\rho \neq 1$, the curve is bowed towards the top left due to covariance term (lower σ_C for the same μ_C, compared to $\rho = 1$ case). For example, look at the midpoint of the line joining A and B.

Question: Can we find the minimum variance portfolio?

This is an optimization problem:

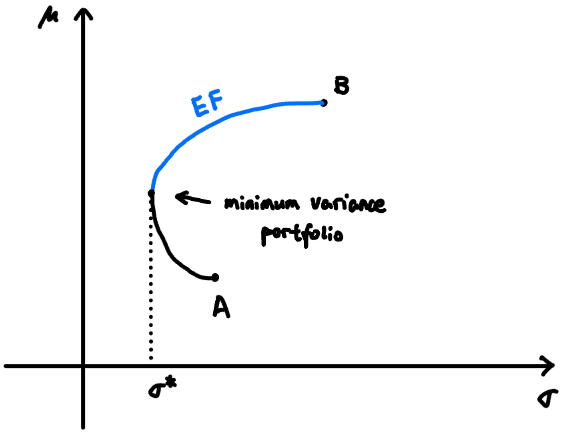
$$\min_{w_A, w_B} w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \rho\sigma_A\sigma_B \text{ s.t. } w_A + w_B = 1.$$

Taking derivatives and setting to 0, we get

$$w_A^* = \frac{\sigma_B^2 - \rho\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}, \quad w_B^* = 1 - w_A^*.$$

Definition 3.8 (**Efficient Frontier**).

The **Efficient Frontier** is the set of optimal portfolios, i.e. a portfolio is optimal if no other portfolio gives higher expected return AND lower risk at the same time.



All economic agents will be on the Efficient Frontier.

Example. Let $\mu_A = 8\%, \sigma_A = 12\%, \mu_B = 14\%, \sigma_B = 20\%, \rho = 0.3$. Let $w_A = 0.8, w_B = 0.2$. Then,

$$\mu_C = 0.8 \cdot 8\% + 0.2 \cdot 14\% = 9.2\%.$$

Also,

$$\sigma_C = \sqrt{0.8^2 \cdot 12\%^2 + 0.2^2 \cdot 20\%^2 + 2 \cdot 0.8 \cdot 0.2 \cdot 0.3 \cdot 12\% \cdot 20\%} = 11.4\%.$$

There are two sources of risk!

- Systematic risk.
- Idiosyncratic risk (firm-specific risk).

Lecture 9, 2025/10/01

Question: What should be the optimum portfolio for a risk averse agent?

Case 3: There are n risky assets, i.e. $A_1, A_2, \ldots, A_n$. The mean vector is

$$\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_n)^\top.$$

The variance-covariance matrix is

$$\Sigma_{n\times n} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$$

where $\sigma_{ij} = \mathrm{Cov}\left(A_i, A_j\right)$ and $\sigma_{ii} = \mathrm{Var}\left(A_i\right) = \sigma_i^2$.

Proposition 3.3 (**Properties of** Σ).

(1) Σ is symmetric.

(2) There are no zero rows in Σ (i.e. no risk-free assets).

(3) Σ is invertible.

For $n = 2$, we have

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}.$$

Let $C = (w_1, w_2, \ldots, w_n)$ be a portfolio such that $\sum_{i=1}^n w_i = 1$, i.e. $e^\top \mathbf{w} = 1$ where $e = (1, 1, \ldots, 1)^\top$ and $\mathbf{w} = (w_1, w_2, \ldots, w_n)^\top$.

Feasible Set

Question: What are the different μ and σ we can generate using these n assets?

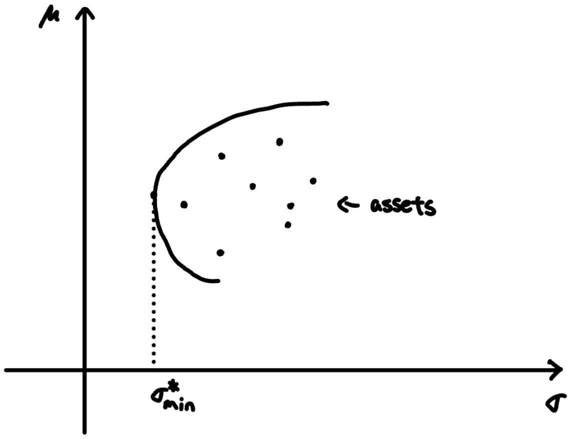
We have $C = w_1 A_1 + w_2 A_2 + \cdots + w_n A_n$. Then,

$$\begin{aligned}
\mu_C &= w_1\mu_1 + w_2\mu_2 + \cdots + w_n\mu_n = \mathbf{w}^\top \boldsymbol{\mu}, \\
\sigma_C^2 &= \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \operatorname{Cov}\left(A_i, A_j\right) \\
&= w_1^2\sigma_{11} + w_2^2\sigma_{22} + \cdots + w_n^2\sigma_{nn} + 2\sum_{i<j} w_i w_j \sigma_{ij} \\
&= \mathbf{w}^\top \Sigma \mathbf{w}.
\end{aligned}$$

Check variance for $n = 2$:

$$\sigma_C^2 = \begin{pmatrix} w_1 & w_2 \end{pmatrix} \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = w_1^2\sigma_{11} + w_2^2\sigma_{22} + 2w_1w_2\sigma_{12}.$$

Question: What does the feasible set look like?



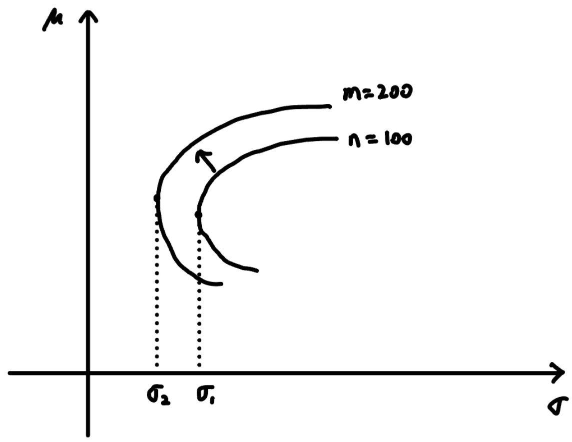
The feasible set is non-linear in general, and is bowed towards the top-left.

Remark (**Summary**).

- There is a value to diversification, i.e. σ_C^2 can be less than the weighted average of individual variances.
- There are limits to diversification.

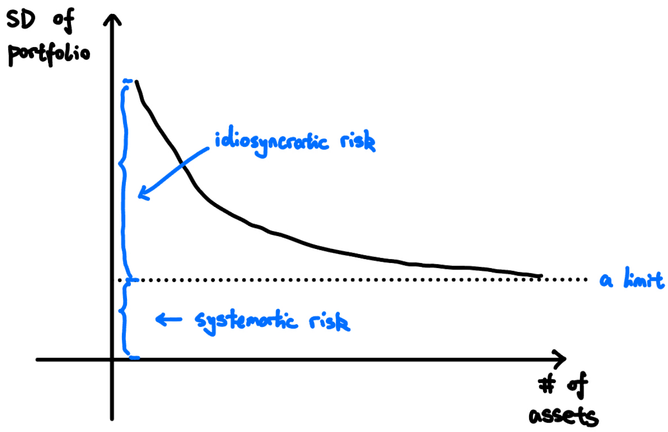
  It is possible that all assets $A_1, \ldots, A_n$ lie below the feasible set.

As n increases, the feasible set moves towards the top-left:



There are two kinds of risks:

- Idiosyncratic risk (firm-specific risk).
- Systematic risk or market risk (affect all firms).



A well-diversified portfolio can get rid of all idiosyncratic risk, but not the systematic risk.

What matters are not the variances, but the covariances. That is, high-variance assets that have low covariances with other assets can reduce the overall portfolio risk, and vice versa.

Example. Suppose there are n assets each with the same standard deviation σ and the same correlation ρ (between any two assets). Consider the equally-weighted portfolio

$$w^\top = \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right) = \text{equally-weighted portfolio.}$$

What is the variance of this portfolio?

Proof. We have

$$\begin{aligned}
\text{Var} &= \underbrace{\left(\frac{1}{n}\right)^2 \sigma^2 + \left(\frac{1}{n}\right)^2 \sigma^2 + \cdots + \left(\frac{1}{n}\right)^2 \sigma^2}_{n \text{ terms}} + n(n-1)\left(\frac{1}{n}\right)^2 \rho\sigma^2 \\
&= \frac{\sigma^2}{n} + \frac{n(n-1)}{n^2}\rho\sigma^2 \\
&= \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2.
\end{aligned}$$

□

Note that as $n \to \infty$, we have $\frac{\sigma^2}{n} \to 0$, whereas $\frac{n-1}{n}\rho\sigma^2 \not\to 0$. So, covariance effect will remain.

Some Technical Points

Here are some questions:

(1) Find the minimum variance portfolio for n-asset problem.

Note that for $(w_1, w_2, \dots, w_n)$, the optimization problem is

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \text{ s.t. } e^\top \mathbf{w} = 1.$$

We can use Lagrange multipliers to solve this and get

$$\mathbf{w}^* = \frac{\Sigma^{-1} e}{e^\top \Sigma^{-1} e}.$$

(2) How to find the Efficient Frontier (EF)?

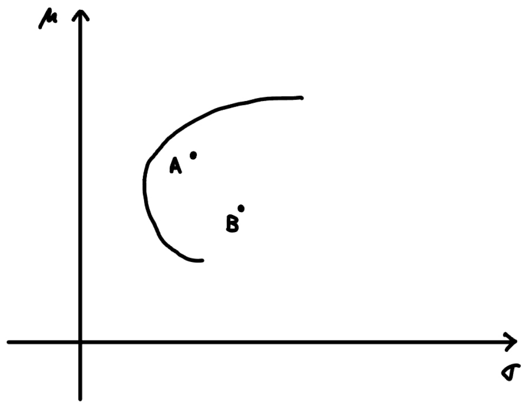
Method 1: Fix σ and maximize μ. That is,

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} \text{ s.t. } \mathbf{w}^\top \Sigma \mathbf{w} = \sigma^2, \quad e^\top \mathbf{w} = 1.$$

Method 2: Fix μ and minimize σ. That is,

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \text{ s.t. } \mathbf{w}^\top \boldsymbol{\mu} = \mu, \quad e^\top \mathbf{w} = 1.$$

Even if asset A dominates asset B (i.e. $\mu_A > \mu_B$ and $\sigma_A < \sigma_B$), it is possible to put positive weights on B because of covariances.



Case 4: The $n+1$ asset model.

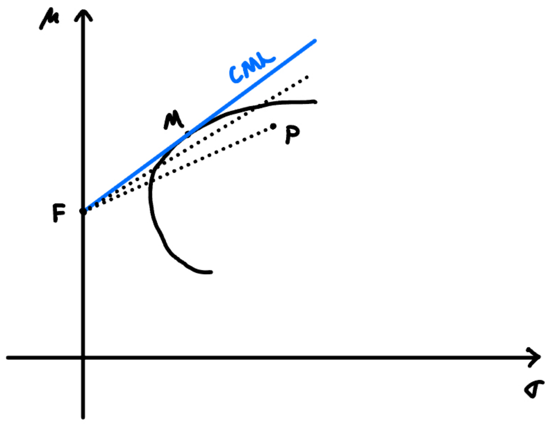
There are n risky assets $A_1, A_2, \dots, A_n$ and one risk-free asset F.

Question: What should be the optimum portfolio?

Answer: First, solve the n risky asset problem, i.e. find the EF for the n risky assets.

Sub-Question: What risky asset should we choose?

Answer: The tangency portfolio M on the steepest CAL (i.e. CML).



All agents (irrespective of their risk aversion) will choose M as their optimum risky portfolio.

Definition 3.9 (**Capital Market Line (CML)**).

The **Capital Market Line (CML)** is the steepest CAL. It also maximizes the Sharpe ratio.

Note. CML is the new efficient frontier.

Question: What is the tangency portfolio M?

Example. Let $n = 2$ and let A and B be the two risky assets. Suppose $M = (0.2, 0.8)$, i.e. $w_A = 0.2$, $w_B = 0.8$. Then, the optimum risky portfolio is 20% in asset A and 80% in asset B.

Note. The tangency portfolio = Market share portfolio.

Example. Suppose there are 3 companies with the following market shares:

- A: \$5 million.
- B: \$3 million.
- C: \$2 million.

Then, $M = (0.5, 0.3, 0.2)$. The optimum risky portfolio is the market portfolio.

The optimum risky portfolio = market portfolio.

Mutual Fund Theorem states that all agents will choose the same risky portfolio M (i.e. the tangency portfolio/market portfolio) due to diversification.

ETF = Exchange Traded Fund, is a good approximation of M.

Passive investment is efficient (i.e. no need to pick stocks, just invest in the market portfolio).

Lecture 10, 2025/10/06

Recall:

(1) The trade-off between a risky asset and a riskless asset is linear (CAL). Agents will choose a point on the CAL (depending on A, the coefficient of risk aversion).

(2) The trade-off between risky assets (n-asset model) is non-linear, in general. The reason for this is the covariances.

Main points:

- Value of diversification.
- Limits to diversification (you cannot get rid of the systematic risk).

Efficiency frontier (EF): all agents (in the n-risky asset model) will be on the EF.

Note. On the EF, to increase μ, you have to increase σ.

(3) The $n+1$ asset model.

$M =$ "best" risky portfolio = tangency portfolio.

CML = Capital Market Line. The slope of the CML is the Sharpe Ratio of M.

All agents will be on the CML.

Tangency Portfolio:

The tangency portfolio = market portfolio = $(w_1, \ldots, w_n)$ where w_i = market share for firm i $= \frac{p_i x n_i}{\sum_j \mu_j n_j}$.

An ETF (Exchange Traded Fund) is a good approximation of M.

3.3 Security Market Line (SML)

Theorem 3.4 (Capital Asset Pricing Model (CAPM)).
Take any asset i in M. The CAPM equation is

$$\mu_i = r_f + \beta_i(\mu_M - r_f)$$

where

- μ_i = expected return of asset i (in percentage).
- r_f = fixed (risk-free) rate.
- μ_M = expected return of the market.
- $\beta_i = \frac{\mathrm{Cov}(A_i, M)}{\sigma_M^2}$ (measure of systematic risk of asset i).

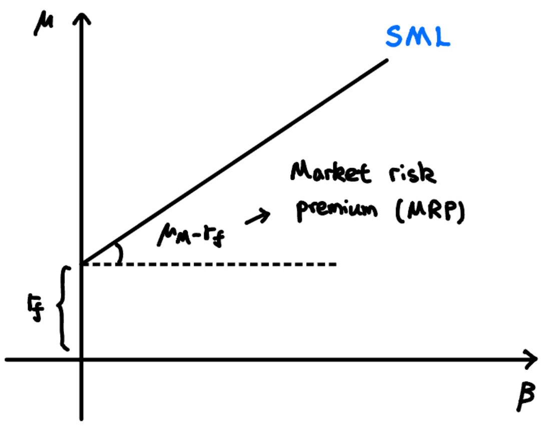
The "right" measure of risk is beta, whereas σ is the volatility.

Remark. The right interest rate for asset i is what you would have made on a similar asset (i.e. an asset with the same beta), which is given by the CAPM equation.

Definition 3.10 (**Security Market Line (SML)**).

The **Security Market Line (SML)** is the graph of the CAPM equation in the (β, μ)-plane:

$$\mu_i = r_f + \beta_i(\mu_M - r_f).$$



The SML gives us the expected return of an asset, given its beta. Note that

$$\beta_i = \frac{\operatorname{Cov}(A_i, M)}{\sigma_M^2} = \text{ how sensitive is asset } i \text{ to changes in the market.}$$

- $\beta = 1 \implies$ the asset is "as sensitive" as the market.
- $\beta > 1 \implies$ the asset is "more sensitive" than the market.
- $\beta < 1 \implies$ the asset is "less sensitive" than the market.

Example. Consider the project: $C_0 = -100$, $C_1 = 110$. Suppose $r_f = 2\%$, $\mu_M = 7\%$, $\beta = 1.1$. What is the "right" interest rate?

Proof. The average return of a "similar" project is

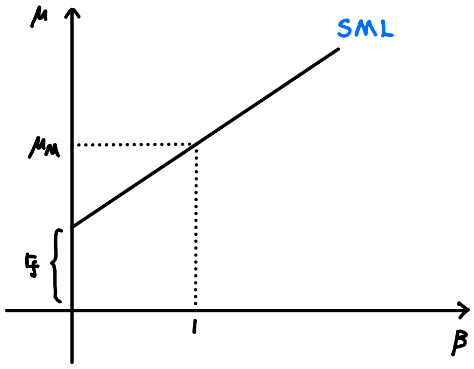
$$\mu_i = r_f + \beta_i(\mu_M - r_f) = 2\% + 1.1 \times (7\% - 2\%) = 7.5\%.$$

The interest rate for the project = expected value of a "similar" project (similar project means the same beta). □

Question: Suppose $\beta = 1$. What should be the return?

Note that

$$\mu_i = r_f + 1 \cdot (\mu_M - r_f) = \mu_M.$$



If $\beta > 1 \implies \mu_i > \mu_M$ and if $\beta < 1 \implies \mu_i < \mu_M$.

If $\beta = 0$, then we have a risk-free asset. Note that

$$\mu_i = r_f + 0 \cdot (\mu_M - r_f) = r_f.$$

Question: Can we have $\beta < 0$?

Answer: Not sure.

Question: Suppose asset A has β_1 and asset B has β_2. Let $C = w_1 A + w_2 B$. What is β_C?

Proof. Given that

$$\beta_A = \frac{\operatorname{Cov}(A, M)}{\sigma_M^2}, \quad \beta_B = \frac{\operatorname{Cov}(B, M)}{\sigma_M^2}.$$

For C, we have

$$\begin{aligned} w_1 \beta_A + w_2 \beta_B &= w_1 \frac{\operatorname{Cov}(A, M)}{\sigma_M^2} + w_2 \frac{\operatorname{Cov}(B, M)}{\sigma_M^2} \\ &= \frac{w_1 \operatorname{Cov}(A, M) + w_2 \operatorname{Cov}(B, M)}{\sigma_M^2} \\ &= \frac{\operatorname{Cov}(w_1 A + w_2 B, M)}{\sigma_M^2} \\ &= \frac{\operatorname{Cov}(C, M)}{\sigma_M^2} = \beta_C. \end{aligned}$$

□

Example.

- Beta for food: low. So expected return is low.
- Beta for cryptocurrency: high. So expected return is high.

Question: How to find betas?

From the SML,

$$\mu_i = r_f + \beta_i(\mu_M - r_f)$$
$$\underbrace{(\mu_i - r_f)}_{y_i} = \beta_i\underbrace{(\mu_M - r_f)}_{x_i}$$

Collect (x_i, y_i) for $i = 1, 2, \ldots, n$ and run a regression. Then,

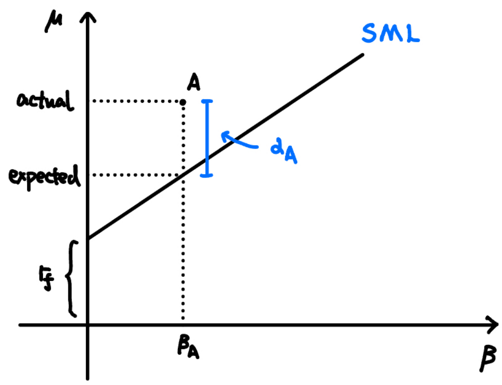
$$\hat{\beta} = \text{ the estimate for } \beta.$$

Performance Measures

Definition 3.11 (Jensen's Alpha).
The **Jensen's Alpha** of asset i is defined by

$$\alpha_i = \text{actual return of asset } i - \text{return predicted by SML}$$

- If $\alpha > 0$, then the asset is underpriced.
- If $\alpha < 0$, then the asset is overpriced.



Definition 3.12 (**Treynor Ratio**).

The **Treynor Ratio** of asset P is defined by

$$\mathrm{TR} = \frac{\mu_P - r_f}{\beta_P}.$$

If the asset lies on the SML, then $\mathrm{TR} = \mu_M - r_f$.

Example. XYZ is an all-equity firm with $\beta = 1.21$. Let MRP $= 4\%$, $r_f = 6\%$, where MRP is the market risk premium, i.e. $\mu_M - r_f$. Let $C_0 = -100$ and

Projects	C_1
A	140
B	120
C	109

What projects should we choose?

Proof. The "right" interest rate is

$$\mu_i = r_f + \beta_i(\mu_M - r_f) = 6\% + 1.21 \times 4\% = 10.84\%.$$

So A and B are profitable projects, whereas C is not. □

Lecture 11, 2025/10/08

Recap:

The expected return of an asset depends on the following:

(1) r_f = risk-free rate.

(2) MRP = market risk premium = $\mu_M - r_f$.

(3) β_i = how sensitive your asset is to the market (appropriate measure of risk).

Note that MRP is the "common factor" for all assets and β_i is an appropriate measure of risk.

Proposition 3.5 (**Properties of Beta**).

(1) $\beta_i = 1 \implies \mu_i = \mu_M$.

(2) $\beta_i > 1 \implies \mu_i > \mu_M$.

(3) $\beta_i < 1 \implies \mu_i < \mu_M$.

(4) $\beta_i = 0 \implies \mu_i = r_f$.

Suppose asset A has β_1 and asset B has β_2. Let $C = w_1 A + w_2 B$. Then,

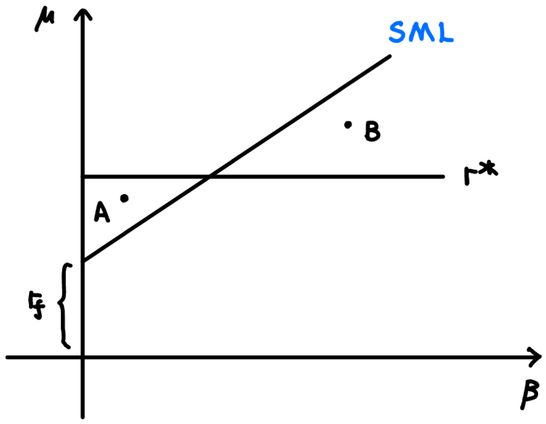
$$\beta_C = w_1 \beta_A + w_2 \beta_B.$$

Definition 3.13 (**Hurdle Rate**).

The **Hurdle Rate** is the minimum acceptable rate of return on an investment.

Note. That is, if the expected return of a project is $> r^*$, then accept the project; otherwise, reject.

Some companies have a fixed hurdle rate r^*, which can lead to bad investment decisions.



Note that project A is above the SML, so it is a good project. However, the hurdle rate rule suggests rejecting it. Also, project B is below the SML, but the hurdle rate rule suggests accepting it.

Example. Suppose a 2 risky asset model with $\rho = 0$ and the following data:

	# of shares	PPS	μ	σ
A	400	50	8%	12%
B	1000	80	14%	20%

(1) Find the minimum variance portfolio.

Proof. We have

$$w_A^* = \frac{\sigma_B^2 - \rho\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B} = \frac{20\%^2 - 0}{12\%^2 + 20\%^2 - 0} = \frac{400}{544} = 0.7353,$$
$$w_B^* = 1 - w_A^* = 0.2647.$$

So, the minimum variance portfolio is $\begin{pmatrix} 0.7353 \\ 0.2647 \end{pmatrix}$. □

Note that the optimization problem here is

$$\min_{w_A} w_A^2\sigma_A^2 + (1 - w_A)^2\sigma_B^2.$$

(2) What is M?

Proof. Company A's value is $400 \times 50 = 20000$. Company B's value is $1000 \times 80 = 80000$. So, A's market share is

$$w_A = \frac{20000}{20000 + 80000} = 0.2,$$

so the market portfolio is $M = (0.2, 0.8)$. □

(3) Find β_A.

Proof. We have

$$\beta_A = \frac{\operatorname{Cov}(A, M)}{\sigma_M^2}.$$

The numerator is

$$\operatorname{Cov}(A, 0.2A + 0.8B) = 0.2\operatorname{Cov}(A, A) + 0.8\operatorname{Cov}(A, B) = 0.2\sigma_A^2 + 0.8 \cdot 0.$$

Plug into the formula to get β_A. □

(4) Find β_B using an alternative method.

Proof. Note that $\beta_M = 1$ and $M = 0.2A + 0.8B$. So,

$$\beta_M = 0.2\beta_A + 0.8\beta_B = 1 \implies \beta_B = \frac{1 - 0.2\beta_A}{0.8}.$$

We can solve for β_B using part (3). □

(5) Find r_f.

Proof. Create asset C such that $\beta_C = 0$:

$$C = w_A A + w_B B.$$

We must have the following:

$$\begin{cases} w_A + w_B = 1 \\ \beta_C = w_A\beta_A + w_B\beta_B = 0 \end{cases}$$

Solve for $\hat{w}_A, \hat{w}_B$, then

$$r_f = \mu_C = \hat{w}_A\mu_A + \hat{w}_B\mu_B.$$

□

4 Arbritrage Pricing Theory (APT)

4.1 APT and Factor Models

From CAPM,

$$\mu = r_f + \beta \cdot \text{common factor}.$$

Instead of one common factor, we allow for k common factors.

- $k = 1 \implies$ single factor model.
- $k > 1 \implies$ multi-factor model.
- $k = 1$ and the common factor is the market: market model.

Assumptions of APT

A1 Individual assets assumption: there are n firms, the actual return of asset i is

$$R_i = \mu_i + m_i + \epsilon_i,$$

where

- R_i is the actual return of asset i.
- μ_i is the expected return of asset i.
- m_i is the set of common factors.
- ϵ_i is the firm-specific individual factor (i.e. idiosyncratic error).

We have

$$m_i = \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k.$$

There are k common factors $F_1, F_2, \dots, F_k$ with β_{ij} = sensitivity of ith firm to the factor j.

Example. The firm is Air Canada. The common factors are

$$m_i = \begin{cases} \text{Oil prices.} \\ \text{GDP of the world.} \\ \text{Inflation.} \end{cases}$$

And ϵ_i = pilots going on strike.

For $k = 1$,

$$R_i = \mu_i + \beta_i F + \epsilon_i.$$

We also assume the following:

- $\mathbb{E}\left[F_j\right] = 0 \ \forall j = 1, \dots, k$, where F is the **surprise factor**.
- $\mathbb{E}\left[\epsilon_i\right] = 0$ and idiosyncratic errors are independent of each other, i.e. $\mathrm{Cov}\left(\epsilon_i, \epsilon_j\right) = 0$ for $i \neq j$.
- $\mathrm{Cov}\left(F_j, F_{j'}\right) = 0$ for $j \neq j'$ and $\mathrm{Cov}\left(F_j, \epsilon_i\right) = 0 \ \forall i, j$.

A2 Portfolio assumptions: a well-diversified portfolio has $\epsilon = 0$.

Example. For $k = 1$, a well-diversified portfolio has

$$\begin{cases} R_1 = \mu_1 + \beta_1 F + \epsilon_1, \\ R_2 = \mu_2 + \beta_2 F + \epsilon_2, \\ \vdots \\ R_n = \mu_n + \beta_n F + \epsilon_n. \end{cases}$$

Consider the portfolio C with weights $w_1, w_2, \dots, w_n$:

$$C = w_1 R_1 + w_2 R_2 + \cdots + w_n R_n.$$

Then,

$$R_c = \mu_c + \beta_c F + \underbrace{(w_1 \epsilon_1 + w_2 \epsilon_2 + \cdots + w_n \epsilon_n)}_{=0}$$

where $\mu_c = w_1 \mu_1 + w_2 \mu_2 + \cdots + w_n \mu_n$ and $\beta_c = w_1 \beta_1 + w_2 \beta_2 + \cdots + w_n \beta_n$.

A3 No arbitrage assumption: if two assets give you the same future returns, they must cost the same today.

Theorem 4.1. Under assumptions A1, A2, and A3, take any asset i,

$$\mu_i = r_f + \beta_{i1} \gamma_1 + \cdots + \beta_{ik} \gamma_k,$$

Note. $k = 1 \implies \mu_i = r_f + \beta_i \gamma$ (i.e. CAPM). Take $\gamma = M$, Then

$$\mu_i = r_f + \beta_i M.$$

Since $M = \mu_M - r_f$, we have $\mu_i = r_f + \beta_i(\mu_M - r_f)$.

Lecture 12, 2025/10/20

Recall:

- CAPM: $\mu_i = r_f + \beta_i \cdot \text{MRP}$, where $\text{MRP} = \mu_M - r_f$ is the systematic factor.
- APT: instead of one systematic factor, we have multiple systematic factors, i.e. multi-factor model.
- APT Assumptions:
  - A1: Take any asset i,

    $$R_i = \mu_i + \underbrace{m_i + \epsilon_i}_{\text{noise}},$$

    Noise has two components:

    * $m_i = \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k$ where β_{ij} = sensitivity of the ith asset to the jth factor and F_j = common factor j.

      Example. Let F_j = oil prices, i = Ford, and i' = Tesla. Note that betas can act in the opposite direction for the same factor.

    * ϵ_i = idiosyncratic factor = firm-specific noise.

    We assume

    * $\mathbb{E}\left[F_j\right] = 0 \; \forall j$.
    * $\mathbb{E}\left[\epsilon_i\right] = 0 \; \forall i$ and $\perp\!\!\!\perp (\epsilon_i, \epsilon_j) \implies \operatorname{Cov}\left(\epsilon_i, \epsilon_j\right) = 0$ for $i \neq j$.
    * $\operatorname{Cov}\left(F_j, F_{j'}\right) = 0 \; \forall j \neq j'$ and $\operatorname{Cov}\left(F_j, \epsilon_i\right) = 0 \; \forall i, j$.
  - A2: Take any well-diversified portfolio i. In such a portfolio, $\epsilon = 0$. Thus,

    $$R_i = \mu_i + m_i.$$

- A3: No arbitrage assumption. If two assets give you the same payoff, the price must be the same today.

Summary of A1 and A2:

- For individual assets: $R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k + \epsilon_i$.
- For portfolios: $R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k$.

Special Cases

- $k = 1$ (single factor model) for portfolios: $R_i = \mu_i + \beta_i F$.
- $k = 1$ and $F = M$ (market factor model) for portfolios: $R_i = \mu_i + \beta_i M$. Then, $\mathbb{E}[R_i] = \mu_i$ and $\mathrm{Var}(R_i) = \beta_i^2 \sigma_M^2$.

Recall Theorem 4.1: Under assumptions A1, A2, and A3, take any asset i,

$$\mu_i = r_f + \beta_{i1}\gamma_1 + \cdots + \beta_{ik}\gamma_k.$$

Special case: for $k = 1$, $\mu_i = r_f + \beta_i \gamma$.

Proof of Theorem 4.1.
We prove the case $k = 1$. Let us take two portfolios A and B. Then,

$$R_A = \mu_A + \beta_A F,$$
$$R_B = \mu_B + \beta_B F.$$

Question: Can we create an asset C such that C is risk-free?

We need to find weights w_A, w_B such that

$$w_A + w_B = 1, \quad w_A\beta_A + w_B\beta_B = 0.$$

Solving this, we have

$$w_A = \frac{\beta_B}{\beta_B - \beta_A}, \quad w_B = \frac{-\beta_A}{\beta_B - \beta_A}.$$

By no-arbitrage assumption, this asset C should give us $\mu_C = r_f$. So,

$$r_f = w_A\mu_A + w_B\mu_B.$$

Then,

$$r_f = \frac{\beta_B}{\beta_B - \beta_A}\mu_A + \frac{-\beta_A}{\beta_B - \beta_A}\mu_B \implies \frac{\mu_A - r_f}{\beta_A} = \frac{\mu_B - r_f}{\beta_B} = \gamma.$$

Thus,

$$\mu_i = r_f + \beta_i \gamma \quad \text{for } i = A, B.$$

□

Market Model

We have

$$\mu_i = r_f + \beta_i M$$

for all portfolios. Here,

$$\mu_M = r_f + \beta_M M, \quad M = \mu_M - r_f.$$

So, $\mu_i = r_f + \beta_i(\mu_M - r_f)$.

Example. Suppose we have a two factor model. We have

$$R_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \epsilon_i.$$

Let A, B, and C be three well-diversified portfolios with the following.

Portfolio i	β_{i1}	β_{i2}	μ_i
A	1.0	1.5	10%
B	0.5	1.0	7%
C	0.75	1.25	9%

Can we create a portfolio D using just A and B such that D is independent of factor 1?

Proof. We find weights w_A, w_B such that

$$w_A + w_B = 1, \quad w_A\beta_{A1} + w_B\beta_{B1} = 0.$$

□

What is the risk-free rate implied in this model?

Proof. Let w_A, w_B, w_C be the weights for a risk-free portfolio. We have

$$\begin{cases} w_A + w_B + w_C = 1, \\ 1.0w_A + 0.5w_B + 0.75w_C = 0, \\ 1.5w_A + 1.0w_B + 1.25w_C = 0. \end{cases}$$

We can solve for w_A, w_B, w_C. Then, the return of this portfolio (i.e. the risk-free rate) is

$$w_A\mu_A + w_B\mu_B + w_C\mu_C = r_f.$$

□

Note. APT is more practical than CAPM since the number of variables that need to be estimated under APT is less than that under CAPM.

4.2 Factors Affecting Beta

We have the following factors affecting beta:

- Cyclicality of revenues (how sensitive stocks are to business cycles).
- Operating leverage.
- Financial leverage.

Cyclicality of Revenues

An asset that is more sensitive to business cycles will have a higher beta.

Example.

- Groceries: low beta.
- Restaurants: higher beta.
- Luxury goods: high beta.

Operating Leverage

Definition 4.1 (Operating Leverage).

Operating leverage is defined by

$$\text{OL} = \frac{\%\text{ change in earnings}}{\%\text{ change in sales}}.$$

Higher operating leverage $\implies$ higher beta.

Example. The following information for a project is given.

- Revenue: \$250 per year forever.
- Cost: \$50 per year forever.
- Assume CAPM holds and $r_f = 5\%$, $\mathbb{E}[r_M] = 10\%$, $\beta = 0.6$.

What is the NPV of the project?

Proof. We find the "right" interest rate first.

$$\mu = r_f + \beta(\mathbb{E}[r_M] - r_f) = 0.05 + 0.6(0.10 - 0.05) = 0.08.$$

Then,

$$NPV = \frac{250}{0.08} - \frac{50}{0.08} = \$2500.$$

□

Suppose the cost was fixed. Then, what is the NPV of the project?

Proof. The "right" interest rate for revenues is 8%. The "right" interest rate for costs is $r_f = 5\%$.

Then,

$$NPV = \frac{250}{0.08} - \frac{50}{0.05} = \$2125.$$

□

Case 1: The project is less sensitive to sales.

Case 2: The project is more sensitive to sales.

Example (**Continued**).

What is the beta of the project where the cost is fixed?

Proof. Recall that

$$NPV = NPV_{\text{revenues}} + NPV_{\text{costs}} = \frac{250}{0.08} - \frac{50}{0.05}.$$

Then,

$$w_{\text{revenues}} = \frac{3125}{2125} = 1.47, \quad w_{\text{costs}} = \frac{-1000}{2125} = -0.47.$$

Thus,

$$\beta_{\text{project}} = 1.47 \cdot 0.6 + (-0.47) \cdot 0 = 0.88.$$

□

Financial Leverage

So far, we assumed all equity firms.

Question: What is the right interest rate if a firm has debt?

Lecture 13, 2025/10/22

Information Asymmetry

In Finance, this is called agency problems.

Example. In a second-hand car, suppose 50% of cars are lemons (bad cars) and 50% are good cars. A lemon is worth $1000 and a good car is worth $3000. Then, the buyer should only pay for $1000 and accept the fact that they might get a bad car.

Example. Suppose a firm has a cash flow of $\$C$ in perpetuity. So,

$$NPV = \frac{C}{r}$$

where r is the cost of capital.

Liquidity

Higher liquidity $\implies$ easier to trade $\implies$ lower cost of capital.

Definition 4.2 (**Bid-Ask Spread**).

The **bid-ask spread** is defined by

$$\text{Bid-Ask Spread} = \text{Buying price} - \text{Selling price}.$$

Note. A bid-ask spread is higher if there is information asymmetry.

Financial Leverage

So far, we assumed that all firms are all equity. Now, we will allow for debt.

Question: What is the right interest rate if a firm has debt?

Properties of Equity Contracts

- Ownership contract.
- Residual claim.
- Limited liability.

Properties of Debt Contracts (Loan Contracts)

- Interest you pay is tax-deductible.
- Failure to pay interest may lead to bankruptcy.

Remark. Value of firm $V = D + E$ where D = market value of debt, E = market value of equity.

Note. Levered firms: $D > 0$ and unlevered firms: $D = 0$.

Questions:

(1) What is the optimum debt-equity ratio?

Answer: We will answer this later (Modigliani-Miller Theorems)

(2) Given. the debt-equity ratio, what is the "right" interest rate?

Answer: Weighted Average Cost of Capital (WACC).

4.3 Weighted Average Cost of Capital (WACC)

Case 1: No taxes. The WACC is

$$r_{\text{WACC}} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$$

where we get r_E from the SML, and r_D is the interest rate on loans (debt).

Case 2: With taxes. The WACC is

$$r_{\text{WACC}} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D (1 - T_c),$$

where T_c is the tax rate.

Example. Suppose $D = \$1$ million, $r_D = 8\%$, and tax rate $= 40\%$. Then,

$$r_D(1 - T_c) = 8\% \cdot (1 - 0.4) = 4.8\%.$$

Interest paid is $\$80,000$ and Tax break $= 0.4 \cdot 80,000 = \$32,000$. So,

$$\text{Effective interest} = 80,000 - 32,000 = 48,000.$$

Note. Interest is tax-deductible.

Example. A firm has the following information.

- Debt-equity ratio is 0.6.
- Cost of debt is 15.15% (i.e. r_D).
- Cost of equity is 20% (i.e. r_E).
- Tax rate is 34% (i.e. T_c).

Is this a good project?

Proof. The WACC is

$$r_{\text{WACC}} = \frac{D}{D+E} r_D (1 - T_c) + \frac{E}{D+E} r_E.$$

Note that $D = 0.6E$. Then, we can compute to get $r_{\text{WACC}} = 16.25\%$. Then, plug r_{WACC} (as the interest rate) into NPV formula to see if the project is good. □

Example. XYZ is a firm with the following information.

- Common shares: # of shares = 8.5 million, PPS = \$34, $\beta = 1.2$.
- Bonds: 200000 outstanding bonds ($F = 1000$), 15 year bonds, 7.5% semi-annual coupons are selling for 93% of par.
- Tax rate = 35%, MRP = 7%, $r_f = 5\%$.

Find r_{WACC}.

Proof. We have

$$r_{\text{WACC}} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D (1 - T_c).$$

Note that

$$E = \text{\# of shares} \times \text{PPS} = 8.5 \text{ million} \times 34 = 289 \text{ million}.$$

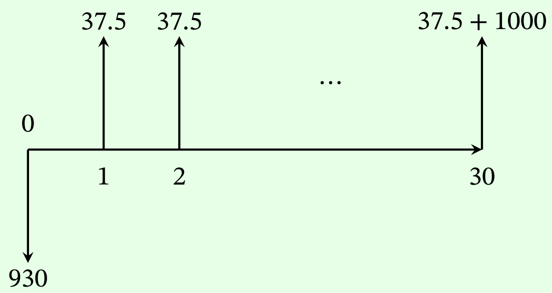
And,

$$D = \text{number of bonds} \times \text{price per bond} = 200,000 \times 930 = 186 \text{ million}.$$

To find r_E, use CAPM:

$$r_E = r_f + \beta \cdot MRP = 5\% + 1.2 \cdot 7\% = 13.4\%.$$

To find r_D, the cash flow of the bond is



Then, equate the NPV to 0 and solve for r_D:

$$0 = -930 + \sum_{t=1}^{30} \frac{37.5}{(1+r_D)^t} + \frac{1000}{(1+r_D)^{30}}.$$

Thus, we can compute to get $r_{\text{WACC}} = 10.27\%$. □

5 Efficient Market Hypothesis (EMH)

Lecture 14, 2025/10/27

Question: What are the conditions for a "good" stock market?

Sub-question: Should the price movements be predictable or unpredictable?

Answer: Unpredictable. Every stock market should have this feature!

EMH Assumptions:

(1) Price movements are unpredictable.

(2) The price of a stock should reflect all relevant information about the stock.

(3) The price adjustment should be instantaneous.

Three Types of EMH:

(1) Weak form.

(2) Semi-strong form.

(3) Strong form.

Definition 5.1 (Weak Form).
The price movement of a stock tomorrow is independent of the price today. That is,

$$P_t = P_{t-1} + \epsilon_t$$

where ϵ_t is a random variable that is independent of P_t's.
This is called the **random walk hypothesis**.

Example (Additional Example: Gambler's Ruin).
Suppose person A and person B have \$$N$ together. Say A has \$$i$ and B has \$$N - i$. They play a game where at each round, A wins \$1 from B with probability $\mathbb{P}(A) = p$ and loses \$1 to B with probability $\mathbb{P}(B) = 1 - p$. The game ends when one person loses all their money. Let $\mathbb{P}(i)$ be the probability that A will win the game if A starts with \$$i$. What is $\mathbb{P}(i)$?

Proof. Note that

$$\mathbb{P}(i) = p \cdot \mathbb{P}(i+1) + (1-p)\mathbb{P}(i-1) \quad \text{for } i = 1, 2, \dots, N-1.$$

Also, $\mathbb{P}(0) = 0$ and $\mathbb{P}(N) = 1$ are the boundary conditions. For example, let $p = \frac{1}{2}$. Then,

$$\mathbb{P}(i) = \frac{i}{N}.$$

Say $N = 200$, $p = 0.49$, $i = 100$. Then, $\mathbb{P}(i) < 0.01$, so don't gamble! $\square$

Technical Analysts

They claim that they can see patterns in stock price movements. Then, the weak form $\implies$ technical analysts are guessing.

Definition 5.2 (Semi-Strong Form).
The price of a stock takes into account all relevant public information.

Note. That is, one cannot beat the market.

Fundamental Analysts

Under the semi-strong form, fundamental analysts are also guessing.

Midterm Review

- MC/Short answer questions.
- 5 Problems,
- Total points: 44.
- Don't spend too much time.
- Don't overthink.

Investment Rules

- Can we calculate NPV, IRR, PI for simple cash flow problems?
- Do we know the weaknesses of IRR, PI, DPP?
- Do we know independent and mutually exclusive projects?
- Do we know when to do incremental analysis (for IRR, PI)?
- Do we know the weaknesses of NPV? (E.g. projects with unequal lives and EAC).

Capital Budgeting

- Do we know how to deal with nominal and real interest rates? (E.g. from one to another, and how to compare cash flows, i.e. be consistent).
- Can we find cash flows from accounting statements, i.e. $(R_t - C_t)(1 - T_c) + DT_c$?
- Do we know what PVCCATS means?
- Do we know the half-year rule?
- Capital budgeting under uncertainty:
  - Decision Trees.
  - Sensitivity analysis, etc.

Utility Theory

- What is the von-Neumann Morgenstern utility function?
- What is risk aversion, risk neutral, risk loving?
- What is certainty equivalent?
- Do we know the degrees of risk aversion?
- Do we know the relationship between the von-Neumann Morgenstern utility function and the mean-variance utility function?

CAPM

- What is the CAL? That is, one risky asset + one riskless asset.
  - What means and variances can we generate?

- What is the amount to invest in the risky asset? That is, $y^* = \frac{\mu_P - r_f}{A\sigma_P^2}$.
- What is the sharpe ratio? That is, the performance indicator.

- Two risky assets and n risky assets models.
  - Can we construct the minimum variance portfolio?
  - M = tangency portfolio = market portfolio (portfolio of market shares).
- The $n+1$ asset model.
  - What is the SML? That is, $\mu = r_f + \beta(\mu_M - r_f)$, which is the right interest rate.
  - What is beta? That is, $\beta_A = \frac{\text{Cov}(A,M)}{\sigma_M^2}$. Also, $\beta_A = 1$ for market portfolio.
- Other measures of performance:
  - Jensen's alpha.
  - Treynor ratio.

APT

- Individual asset assumption: $R_i = \mu_i + m_i + \epsilon_i = \mu_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \cdots + \beta_{ik}F_k + \epsilon_i$. What is the variance of R_i?
- Portfolios: $R_i = \mu_i + m_i$.
- Suppose a two-factor model. Say

Asset	β_{i1}	β_{i2}
A	0.75	−1
B	1.25	0.25

  Can we create portfolios that are independent of certain factors?

WACC

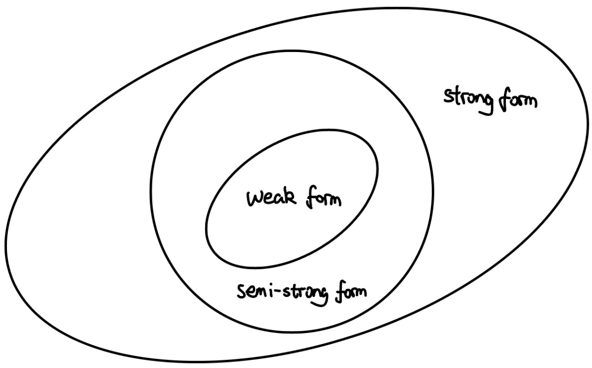
- What is the right interest rate if $D > 0$? That is, $r_{\text{WACC}} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - T_c)$, where r_E is from the SML and r_D is from the YTM of debt.

Lecture 15, 2025/11/03

Note. Today's lecture is definition heavy and may be examined in multiple choice questions.

Recall:

- **Weak form**: you cannot "beat" the market if we know today's prices. We have $P_t = P_{t-1} + \epsilon_t$, e.g. $\epsilon_t \sim N(0, \sigma^2)$. This is called the random walk hypothesis. Sometimes the formula is written as $P_{t+1} = P_t + g + \epsilon_t$ which is called the random walk with drift.
- **Semi-strong form**: you cannot "beat" the market with public information about the asset (e.g. stock).
- **Strong form**: you cannot "beat" the market with all information (public + private) about the asset.



Inefficiencies of the EMH

Technical analysts: they claim that P_{t+1} is based on P_t

Weak form $\implies$ technical analysts are guessing.

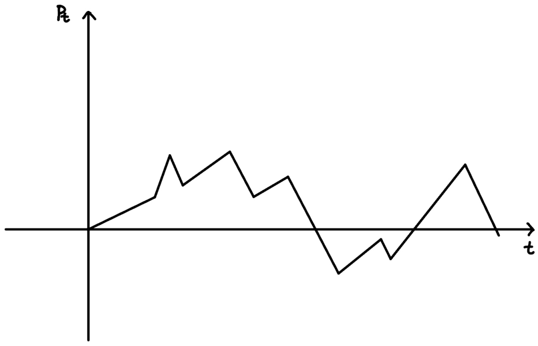
Fundamental analysts: they check whether an asset is over/under priced based on public information.

Semi-strong form $\implies$ fundamental analysts are guessing.

Strong form $\implies$ insider trading is not profitable, where insider trading means illegal trading based on private information.

Evidence For/Against EMH

Question: Can we check whether the weak form is true?

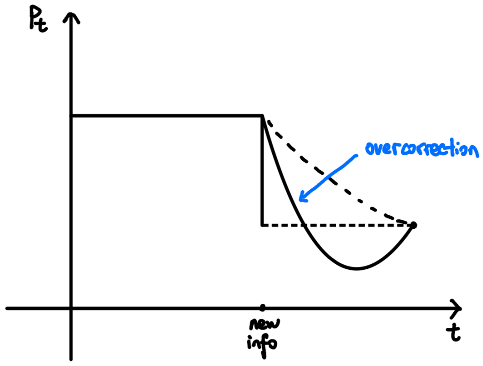


We try to "fit" a random walk to describe the stock price movements. For example, for tossing a fair coin, we can define

$$X_t = \begin{cases} 1 & \text{if H} \\ -1 & \text{if T.} \end{cases}$$

Then, $S_n = \sum_{t=1}^{n} X_t$ is a random walk.

If you can find an investment manager who can consistently beat the market, then the semi-strong form is false.



Absence of Trades: this is an evidence for the EMH, because if all stock prices are correctly priced, then there is no incentive to trade.

Joint Hypothesis Problem: Testing if stocks are correctly priced is an evidence for the EMH, or an evidence for CAPM?

Two Schools of Thought:

- **Classical**: Dutch Book Theorem.
- **Behavioural Finance**: markets are not rational.

6 Capital Structure

6.1 The Basics

Question: Does leverage have any effect on the value of the firm?

Question: What is the "optimum" D/E ratio?

Answer: The Modigliani-Miller Theorem.

Equity

Definition 6.1 (**Equity**).

Equity can be one of the following:

- Ownership contract.
- Residual claims.
- Limited liability.

Definition 6.2 (**Market CAP**).

Market CAP is defined by

$$\text{Market CAP} = \text{Number of shares} \times \text{Price per share}.$$

Types of shares:

- Common shares: 1 share $=$ 1 vote.
- Superior shares: 1 share $>$ 1 vote.
- Limited shares: 1 share $<$ 1 vote.

Shareholders elect the board of directors. The directors then hire management. This is how companies are run.

Voting:

- Non-cumulative voting.
- Cumulative voting.

Example. Suppose A has 75% of shares, B has 25% of shares, and B of directors are to be elected.

- Non-cumulative voting:
  Sequencial elections: A has 75 votes and B has 25 votes. So, A will be able to select
- Cumulative voting: A will have $75 \times 4 = 300$ votes and B will have $25 \times 4 = 100$ votes. Elections are simultaneous: B can allocate all 100 votes to 1 candidate. So, B will be able to elect 1 director.

Cumulative voting $\implies$ Minority shareholders have more power.

Definition 6.3 (**Preferred Shares**).
A **preferred share** specifies the amount of dividend we get every period.

Note. Preferred shares are between equity and debt.

- Like equity: not paying dividends $\not\Rightarrow$ bankruptcy.
- Like debt: pre-specified payments.

We have the following types of preferred shares:

- Fixed preferred shares: the dividend rate is fixed.
- Floating preferred shares: changes with the interest rate (e.g. T-bill rates + 3%).
- Callable preferred shares: the issuer (seller) has a right to call the share back at a pre-specified price.
- Retractible preferred shares: the holder (buyer) has a right to sell the stock back to the issuer at a pre-specified price.
- Redeemable preferred shares: the buyer has the right to redeem the preferred share in exchange for common shares.
- Reset preferred shares: the dividend rate changes (typically every 5 years).

Debt

Definition 6.4 (**Debt**).

Features of debt:

- Loan contract.
- Interest you pay on debt is tax deductible.
- Failure to pay interest $\implies$ bankruptcy.

Two categories of debt:

- Secured debt: backed by an asset (e.g. mortgage).
- Unsecured debt: not backed by an asset. This is also called a **debenture**.

Types of debt:

- Long term debt
- Short term debt
- Trades payable
- Indenture: the contract between the borrower and the lender.

  Example. Restrictions on dividends, hiring good auditors.

- Subordinated debt: your debt is "behind the line" in the event of a bankruptcy.

In the event of a bankruptcy, investors get paid in the following order:

- Senior bond holder
- Junior bond holder
- Debentures
- Trades payable
- Preferred shareholders

6.2 Modigliani Miller Propositions

Definition 6.5 (**Levered Firm**).

A **levered firm** is a firm that has debt, i.e. $D > 0$.

Suppose the firm produces \$$C$/year in perpetuity. Then, the value of the firm is

$$V = \frac{C}{r_{\text{WACC}}}.$$

Recall that

$$r_{\text{WACC}} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - T_c).$$

Question: What D/E ratio should we choose to maximize the firm value (i.e. minimize r_{WACC})?

This brings us to Modigliani and Miller Propositions.

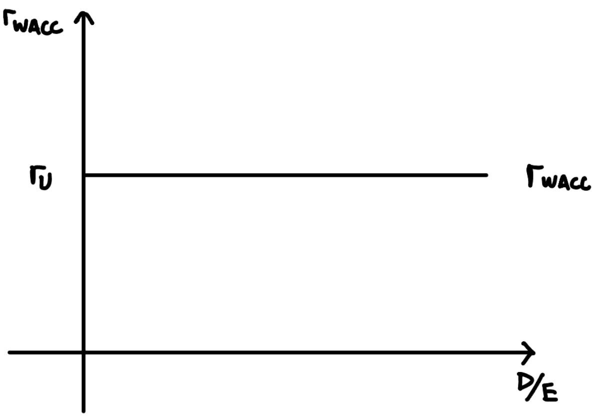
Case 1: No taxes, i.e. $T_c = 0$.

Let V_L = value of the levered firm, and V_U = value of the unlevered firm.

Proposition 6.1 (**M & M Proposition 1 (No Taxes)**).

$V_L = V_U$.

Implications: D/E ratio is irrelevant to the firm value AND r_{WACC} is unaffected by the D/E.



Example. A firm is unlevered (i.e. $D = 0$). The value of the firm is \$8000. There are 400 outstanding shares. So,

$$\text{PPS} = \frac{8000}{400} = \$20.$$

The firm is considering taking debt. They borrow \$4000 and buy off 200 shares, $r_D = 10\%$. What is the D/E ratio after the buyback?

Proof. Before leverage, i.e. unlevered:

$$D = 0, \quad E = 8000.$$

After leverage, i.e. levered:

$$D = 4000, \quad E = 8000 - 200 \times 20 = 4000 \implies \frac{D}{E} = 1.$$

Note that the value of the firm remains the same for both cases. □

Example (No Debt Scenario).

	Recession	Normal	Expansion
Earnings	\$400	\$1200	\$2000
ROA	$\frac{400}{8000} = 5\%$	15%	25%
ROE	5%	15%	25%
EPS	$\frac{400}{400} = \$1$	\$3	\$5

Example (Debt Scenario).

	Recession	Normal	Expansion
Earnings (before interest)	\$400	\$1200	\$2000
Interest	$4000 \times 10\% = \$400$	\$400	\$400
Net Income	0	800	1600
EPS	0	$\frac{800}{200} = \$4$	$\frac{1600}{200} = \$8$

Question: What capital structure is "better"?

Suppose we have \$2000 and we want to invest in the levered company. We expect:

	Recession	Normal	Expansion
Earnings	0	400	800

Suppose the firm remains unlevered. Can we generate the levered payoffs?

Strategy: Borrow \$2000 and use \$2000 of our own money to buy \$4000 of the unlevered firm's shares. Then, we have:

	Recession	Normal	Expansion
Earnings before interest	200	600	1000
Interest	200	200	200
Net Income	0	400	800

Suppose we have \$2000 and we want unlevered payoffs, but the firm becomes levered. We expect:

	Recession	Normal	Expansion
Earnings	100	300	500

Strategy: Invest \$1000 in the levered firm and lend \$1000 at 10% interest. Then, we have:

	Recession	Normal	Expansion
Earnings	0	200	400
Interest	100	100	100
Net Income	100	300	500

The firm is not adding any value by changing the capital structure.

Definition 6.6 (**Homemade Leverage**).
If a firm remains unlevered, but we want levered payoffs, we change our own D/E ratio to the desired level. This is called **homemade leverage**.

Recall that for the no tax case,

$$r_{\text{WACC}} = \frac{E}{V} r_E + \frac{D}{V} r_D.$$

By MM Proposition 1,

$$r_{\text{WACC}} = r_U = \text{unlevered rate of return} \implies r_U = \frac{E}{V} r_E + \frac{D}{V} r_D.$$

Then,

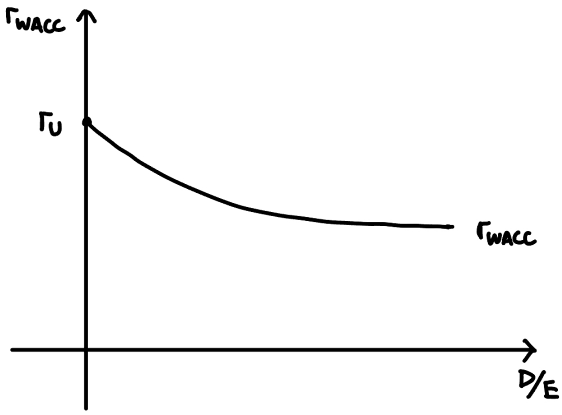
$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

where r_U = cost of capital of the unlevered firm and r_E = cost of equity of the levered firm.

Case 2: With taxes.

Proposition 6.2 (**M & M Proposition 1 (With Taxes)**).
$V_L = V_U + DT_c$, where D = debt, assumed to be in perpetuity.

Note. The value of the firm increases with leverage.



Higher $D/E \implies$ lower $r_{\text{WACC}} \implies$ higher firm value.

Proposition 6.3 (**M & M Proposition 2 (With Taxes)**).

$$r_E = r_U + \frac{D}{E}(r_U - r_D)(1 - T_c).$$

Proof of MM 1 (with taxes).

The firm makes \$EBIT/year in perpetuity. Then, D is also in perpetuity. So,

$$V_U = \frac{\text{EBIT}(1 - T_c)}{r_U}.$$

For the levered, bondholders receive $r_D D$. The shareholders receive $(\text{EBIT} - r_D D)(1 - T_c)$. So,

$$V_L = \text{PV of bondholders} + \text{PV of shareholders}.$$

The total payment is

$$(\text{EBIT} - r_D D)(1 - T_c) + r_D D = \text{EBIT}(1 - T_c) + r_D D T_c.$$

Thus,

$$V_L = \frac{\text{EBIT}(1 - T_c)}{r_U} + \frac{r_D D T_c}{r_D} = V_U + DT_c.$$

□

Remark (**Summary**).

(1) Without taxes: D/E is irrelevant to firm value and r_{WACC} is constant.

(2) With taxes: higher $D \implies$ higher firm value and lower r_{WACC} as D/E increases.

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Recall: We used homemade leverage to prove MM Proposition 1 (no taxes).

Homemade Leverage: create your own D/E ratio to match that of the firm.

The size of the cake is independent of who owns it.

Implications of MM Proposition 1 (with taxes):

- The value of the firm increases with debt.
- r_{WACC} decreases as D/E increases (refer to the plot on the previous page).

Example (**No Tax Case**).

Suppose that ABC and XYZ are two identical firms, except that ABC is unlevered with \$750000 of stocks and XYZ is levered with $D = \$375000$, $E = \$375000$. Also, $r_D = 8\%$, no taxes, and both firms have the same EBIT of \$86000/year in perpetuity.

Q1: Person A owns \$30000 of XYZ's stock. What return is this person expecting?

Proof. Note that EBIT = 86000. Then,

$$\text{Interest paid} = 0.08 \times 375000 = 30000.$$

So,

$$\text{Net earnings} = 86000 - 30000 = 56000.$$

Thus, A's share of the earnings is

$$\frac{56000}{375000} \times 30000 = \$4480.$$

A's rate of return is $\frac{4480}{30000} = 14.93\%$. □

Q2: Can A generate the same return using ABC's stock?

Proof. The strategy is to borrow \$30000 and use A's own \$30000 to buy \$60000 shares of ABC. Then, the earning from ABC is

$$\frac{60000}{750000} \times 86000 = \$6880.$$

Also,

$$\text{Interest on loan} = 0.08 \times 30000 = 2400.$$

So,

$$\text{Net Income} = 6880 - 2400 = 4480.$$

□

Q3: What is r_U for ABC?

Proof. Note that

$$V_U = \frac{86000}{r_U} = 750000 \implies r_U = \frac{86000}{750000} = 11.47\%.$$

□

Q4: What is r_E for XYZ?

Proof. By MM Proposition 2 (no taxes),

$$r_E = r_U + \frac{D}{E}(r_U - r_D) = 11.47\% + 1 \times (11.47\% - 8\%) = 14.93\%. \qquad \square$$

6.3 Limits to Debt

There are three main problems with too much debt:

- Risk of bankruptcy (already included in the stock price).
- Direct costs.
- Agency costs.

Direct Costs

- Lawyers' fees.
- Loss of reputation.

Agency Costs

The interest of the debtholders and that of the management may not be aligned (i.e. conflict of interest).

Example (**Riskier projects may be accepted**).

Suppose $D = 100$.

- Project 1: 50% chance of \$100 and 50% chance of \$200.
- Project 2: 50% chance of \$50 and 50% chance of \$240.

Note that project 1 should be chosen since it has a higher expected return. However, for project 1:

$$\text{Shareholders' payoff} = 0.5 \times 0 + 0.5 \times 100 = 50.$$

For project 2:

$$\text{Shareholders' payoff} = 0.5 \times 0 + 0.5 \times 140 = 70.$$

Even though project 1 is safer, we will choose project 2 since shareholders get a higher expected payoff.

Example (**Profitable Projects may be rejected**).

Suppose $D = 4000$ and suppose the company has two possibilities:

- 50% chance of \$2400.
- 50% chance of \$100.

The available project has $C_0 = -1000$ and $C_1 = 1700$. The project has to be financed through equity.

Without the project, the shareholders' expected payoff is

$$0.5 \times 0 + 0.5 \times 100 = 50.$$

With the project, the shareholders' expected payoff is

$$0.5 \times 100 + 0.5 \times 1800 = 950.$$

However, the initial cost of the project is \$1000 while the income increases by only \$950. So, the company will reject the project.

So having a lot of debt is not a good thing! And we don't know what the optimum D/E ratio is.

Other Theories

- **Free cash flow hypothesis**: more "free" cash $\implies$ more waste $\implies$ lower firm value.

  So having debt $\implies$ interest payments $\implies$ less free cash $\implies$ higher firm value.

- **Signaling theory**: taking debt $\implies$ signal of that the company is doing well.

Remark (**Summary**).

- If the project looks like the assets of the company, use r_{WACC}.
- If the project is financed completely by equity, use r_E (from the SML).
- If the project is financed completely by debt, use r_D, i.e. the YTM of the bonds.
- If the project is risk-free, use r_f, i.e. the T-bill rate.

6.4 Capital Budgeting for Levered Firms

Question: How to calculate NPVs for levered firms?

Method 1: r_{WACC}. Here are the steps:

(1) Find r_{WACC}.

(2) Use C_t and r_{WACC} to calculate NPV.

Method 2: Adjusted Present Value (APV). Splitting the project into an unlevered part and financing effects.

The formula is

$$NPV_{\text{levered}} = NPV_{\text{unlevered}} + DT_c.$$

Remark. The formula above assumes that the debt is in perpetuity. If the debt is not in perpetuity, the second term = PV of tax breaks because of debt.

Example (**APV Method**).

Suppose ABC is a firm with

- $C_0 = -1000000$.
- $D = 300000$ in perpetuity.
- EBIT = 200000 per year in perpetuity.
- $r_U = 10\%$, $T_c = 25\%$, and $r_D = 5\%$.

Find the NPV.

Proof. Note that

$$NPV_{\text{unlevered}} = C_0 + \frac{EBIT(1 - T_c)}{r_U} = -1000000 + \frac{200000 \times 0.75}{0.1} = 500000.$$

Thus,

$$NPV_{\text{levered}} = 500000 + 300000 \times 0.25 = 575000.$$ □

Remark (**Trick**).

If D/E ratio is given, r_{WACC} may be easier to calculate (i.e. use r_{WACC} method). If D is given, use APV. In the example above, we used APV since D is given.

6.5 Dividends

Definition 6.7 (Dividends).
Dividends are periodic payments to the shareholders.

Types of Dividends:

- Cash dividend.
- Stock dividend.

  Example. A 20% stock dividend means:

  $$100 \text{ shares before} \implies 120 \text{ shares now.}$$

  Stock split.

  Example. A $2:1$ stock split means:

  $$50 \text{ shares before} \implies 100 \text{ shares now.}$$

  Reverse stock split.

  Example. A $1:4$ reverse stock split means:

  $$100 \text{ shares before} \implies 25 \text{ shares now.}$$

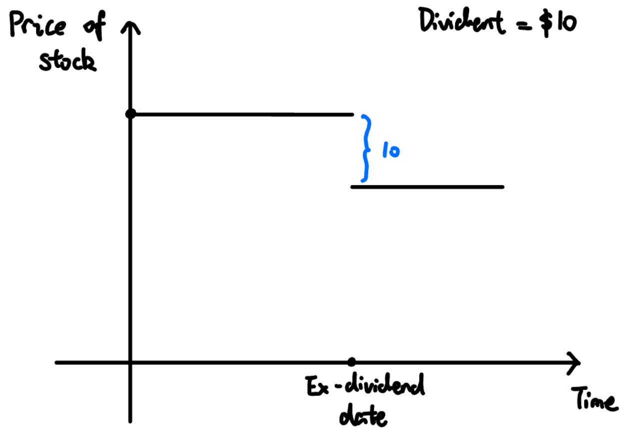
- Stock repurchase.

  Example. The firm buys off shares $\implies$ value of the remaining shares increases.

Important Dates

- Declaration date: the Board of Directors declares the dividend.
- Ex-dividend date: the first day the seller of a share is entitled to the dividend.
- Record date: the date when dividend payment is recorded.
- Payment date: the date when dividend is actually paid.

Ex-dividend date is the only one that matters. The seller who sells the stock on or after the ex-dividend date keeps the dividend.



The stock price will fall by the amount of the dividend on the ex-dividend date.

Question: Does dividend policy matter?

According to the M & M proposition, dividend policies are irrelevant.

Example. Suppose XYZ is a firm that lives for two periods:

Period	Year 0	Year 1
???	1000	1000

Suppose 100 shares in total and the payments are risk-free with $r_f = 10\%$. Let us also assume all dividends are paid out.

Price of stock (before dividends) at year 0:

Current policy: 100% dividend payout.

	Year 0	Year 1
Current Policy	1000	1000
Proposed Policy	1100	890

???

Strategy: sell $10 worth of shares.

???

Clientele Effect: different clients like different dividend policies. For example, older people may prefer regular dividends.

Free Cash Flow Hypothesis:

More dividends paid $\implies$ less "free cash" $\implies$ less waste $\implies$ higher firm value.

Signaling: regular dividend payments is a signal of the company's health.

7 Options

Lecture 19, 2025/11/17

7.1 Intro to Options

Definition 7.1 (**Call Option**).

A **call option** gives the buyer the right, but not an obligation, to buy the asset at a pre-determined price on/before a pre-determined date.

Notations:

- Underlying asset: stock.
- S_0 = price of the stock at time $t = 0$.
- T = expiration date.
- S_T = price of the asset on the expiration date.
- X = pre-agreed price (i.e. strike price).
- Buyer: holder (long position).
- Seller: writer (short position).
- Price of the option = premium.
- "On the day": European option.
- "On/Before expiration date": American option.

Example. Suppose $S_0 = 50$, $T = 1$ year, and $X = 55$ (European). Let V_T be the value of the option on the expiration date. On the day of expiration:

- If $S_T = 53$, then, $V_T = 0$.
- If $S_T = 57$, then, $V_T = 2$.

Note that,

$$V_T = \begin{cases} 0, & S_T < X \\ S_T - X, & S_T \geq X. \end{cases}$$

So, $V_T = \max(0, S_T - X)$.

Missing plots of payoff diagrams for seller and buyer.

Bullish Strategy: The strategy is non-decreasing with the stock price.

Bearish Strategy: The strategy is non-increasing with the stock price.

At the Money

Suppose $t = 0 =$ Nov. 17, $t = T =$ Dec. 31, $S_0 = \$100$, and $X = \$105$.

Definition 7.2 (**In/Out/At the Money**).

If $S_t < X$, the option is out of the money. If $S_t > X$, the option is in the money. If $S_t = X$, the option is at the money.

Definition 7.3 (**Put-Option**).

A **put-option** gives the buyer the right, but not an obligation, to sell an asset at a pre-agreed price X on/before a pre-agreed date T.

Note. This is the opposite of a call option.

Value of Put Option at Expiry

Missing plots of payoff diagrams for seller and buyer.

Example. Suppose ABC is a firm and PPS $= \$100 = S_0$. At the money, call options are getting sold for \$10 each. Suppose $T = 1, X = 100, S_0 = X, r_f = 3\%$.

A person has \$10000 to invest.

Strategy 1: Buy all stocks, which is 100 shares.

Strategy 2: Buy all options, which is 1000 options.

Strategy 3: Put \$9000 in a bank. Buy \$1000 options.

Here are the possible prices at expiry:

	$S_T = 95$	100	105	110	115	120
Strategy 1	9500	10000	10500	11000	11500	12000
Loss/Gain for S1	-5%	0%	5%	10%	15%	20%
Strategy 2	0	0	5000	10000	15000	20000
Loss/Gain for S2	-100%	-100%	-50%	0%	50%	100%
Strategy 3	$9270 + 0$	$9270 + 0$	$9270 + 500$	$9270 + 1000$	$9270 + 1500$	$9270 + 2000$
Loss/Gain for S3	-7.3%					

S3 gives us protection on the downside, while replicating the gain from stock on the upside.

Option Strategies

Protective Put Strategy

- Buy the stock.
- Buy a put-option at a strike price X.

What payoff do we get?

	$S_T < X$	$S_T \geq X$
Long Stock	S_T	S_T
Long Put	$X - S_T$	0
Total Payoff	X	S_T

Missing plots of payoff diagram.

Straddle Strategy

- Buy a call at price X, $T = 1$ year.
- Buy a put at price X, $T = 1$ year.

	$S_T < X$	$S_T \geq X$
Long Call	0	$S_T - X$
Long Put	$X - S_T$	0
Total Payoff	$X - S_T$	$S_T - X$

Missing plots of payoff diagram.

Lecture 20, 2025/11/19

Notation:

- C = price of a call option.
- P = price of a put-option.

Option Strategies

- Protective Put Strategy.
  - Long Stock.
  - Long Put (X).
- Bull-Call-Spread.
  - Buy a call option at strike price X_1.
  - Sell a call option at strike price $X_2 > X_1$.

Example. Suppose $X_1 = 90, X_2 = 100$.

	$S_T < X_1$	$X_1 \leq S_T < X_2$	$S_T \geq X_2$
Long Call (X_1)	0	$S_T - X_1$	$S_T - X_1$
Short Call (X_2)	0	0	$-(S_T - X_2)$
Total Payoff	0	$S_T - X_1$	$X_2 - X_1$

Missing plots of payoff diagram.

- Short Strangle.

  Write an out-of-the-money call option (X_2) and an out-of-the-money put option (X_1), where $X_2 > X_1$.

Example. Suppose $X_1 = 57, S_0 = 60, X_2 = 65$.

	$S_T < X_1$	$X_1 \le S_T < X_2$	$S_T \ge X_2$
Short Put (X_1)	$-(X_1 - S_T)$	0	0
Short Call (X_2)	0	0	$-(S_T - X_2)$
Total Payoff	$S_T - X_1$	0	$X_2 - S_T$

Other Assets that Look Like Options

Missing content.

The value of a firm is $V = D + E$. Then, the value to equity holders is

$$E = \begin{cases} 0, & V < D \\ V - D, & V \ge D. \end{cases} \implies E = \max(0, V - D).$$

The firm is owned by the lenders. The equity holders have a call option on the firm with a strike price $X = D$.

Alternative Way

The equity holders hold the firm. They also have a debt D to repay. The shareholders have a put-option with $X = D$.

	$V < D$	$V \ge D$
Long Firm	V	V
Debt	$-D$	$-D$
Long Put ($X = D$)	$D - V$	0
Total Payoff	0	$V - D$

Definition 7.4 (**Put-Call Parity**).

The **put-call parity** is defined by

$$C + \frac{X}{(1+r_f)^T} = S_0 + P \iff C + Xe^{-r_f T} = S_0 + P$$

where

- S_0 = price of the stock today.
- T = time to expiration.
- X = strike price.
- C = price of the call option (X, T).
- P = price of the put option (X, T).
- r_f = risk-free rate.

Proposition 7.1 (**Law of One Price**).

If two assets give the same future payoffs, then they should cost the same today.

Deriving Put-Call Parity

Strategy 1: Buy a call option, lend $\frac{X}{(1+r_f)^T}$. The net cost is

$$C + \frac{X}{(1+r_f)^T}.$$

	$S_T < X$	$S_T \geq X$
Long Call	0	$S_T - X$
Lent money	X	X
Total Payoff	X	S_T

Strategy 2: Buy the stock, buy a put option at strike price X. The net cost is

$$S_0 + P.$$

	$S_T < X$	$S_T \geq X$
Long Stock	S_T	S_T
Long Put	$X - S_T$	0
Total Payoff	X	S_T

Strategy 1 and 2 give the same payoffs at time T. By the law of one price, they should cost the same today. Thus,

$$C + \frac{X}{(1 + r_f)^T} = S_0 + P.$$

Question: What happens if the put-call parity is violated?

Example. Suppose $T = 1$ year, $S_0 = 110$, $X = 105$, $C = 17$, $P = 5$, $r_f = 5\%$.

Note that $C + \frac{X}{(1+r_f)^T} = 117 \neq 115 = S_0 + P$. Thus, the put-call parity is violated. We can conclude that the LHS (call + bond) is overpriced and the RHS (stock + put) is underpriced.

Strategy: buy low (RHS) and sell high (LHS).

- Buy stock: -110.
- Buy put: -5.
- Sell call: $+17$.
- Borrow money (sell bond): $\frac{105}{1.05} = 100$.

Thus, net cash flow today is $-110 - 5 + 17 + 100 = 2$. We can show that the cash flow at time T is 0 in both cases ($S_T < X$ and $S_T \geq X$), so no risk.

In conclusion, if the put-call parity is violated, we can make a risk-free profit!

7.2 Binomial Option Pricing Model

Example. Suppose $S_0 = 100$, $T = 1$ year. In one year, the stock price can be either uS_0 or dS_0 where $u > d$ and

- u = up-factor.
- d = down-factor.

Suppose $u = 1.2$, $d = 0.9$, $X = 110$, and $r_f = 10\%$. Let C = premium. Then,

- The stock goes up to $uS_0 = 120$ with probability p.
- The stock goes down to $dS_0 = 90$ with probability $1 - p$.

Also,

- If the stock goes up, the stock is worth $C_u = \max(0, uS_0 - X) = 10$.
- If the stock goes down, the stock is worth $C_d = \max(0, dS_0 - X) = 0$.

We want to find C.

Solution 1: Replicating strategy method.

Buy Δ units of stock and lend B dollars to replicate the call option.

- In the up state, the portfolio is worth $\Delta uS_0 + B(1 + r_f)^T = C_u = 10$.
- In the down state, the portfolio is worth $\Delta dS_0 + B(1 + r_f)^T = C_d = 0$.

This gives us the system of equations:

$$\begin{cases} 120\Delta + (1 + 0.1)B = 10 \\ 90\Delta + (1 + 0.1)B = 0 \end{cases}$$

We can solve to get $\Delta = \frac{1}{3}$ and $B = -27.27$.

By the Law of One Price, cost of replicating portfolio = cost of call option. Thus,

$$\text{Cost of replicating strategy} = C = \Delta S_0 + B = \frac{1}{3} \times 100 - 27.27 = 6.06.$$

Solution 2: Delta hedging method.

Step 1, we calculate Δ:

$$\Delta = \frac{\text{Change in the option price}}{\text{Change in the stock price}} = \frac{C_u - C_d}{uS_0 - dS_0} = \frac{10 - 0}{120 - 90} = \frac{1}{3}.$$

Step 2, buy Δ units of the stock and write (sell) 1 option. That is, we buy (go long) $\frac{1}{3}$ of the stock and write (short) 1 option.

	$S_T = 120$	$S_T = 90$
Long $\frac{1}{3}$ stock	40	30
Short 1 option	−10	0
Total Payoff	30	30

This means that this strategy yields \$30 risk-free in $T = 1$ year. Then, the cost of this strategy today is

$$\text{PV of \$30} = \frac{30}{1+r_f} = \Delta S_0 - C \implies C = \Delta S_0 - \frac{30}{1+r_f} = \frac{1}{3} \times 100 - \frac{30}{1.1} = 6.06.$$

Solution 3: Risk-neutral probability method.
The general formula is

$$C = \frac{1}{1+r_f}\left[C_u \underbrace{\left(\frac{(1+r_f)-d}{u-d}\right)}_{q} + C_d \underbrace{\left(\frac{u-(1+r_f)}{u-d}\right)}_{1-q}\right] = \frac{1}{1+r_f}[C_u q + C_d(1-q)].$$

C is independent of p, we can choose any p. If all of the investors are risk-neutral, then C would still be the same!
Imagine a risk-neutral universe.
Fact 1: All assets should give us the same expected return.

$$\text{Return from stock} = \text{Return from the risk-free asset}$$

Let q = probability of the up-factor in this risk-neutral universe. Then,

$$\frac{q \cdot 120 + (1-q) \cdot 90 - 100}{100} = 10\% \implies q = \frac{2}{3}.$$

Fact 2: The value of an asset = PV of expected returns.

$$C = \frac{1}{1+r_f}[C_u q + C_d(1-q)] = \frac{1}{1.1}\left[10 \times \frac{2}{3} + 0 \times \frac{1}{3}\right] = 6.06.$$

Two-Stage Binomial Model

Example. The stock price changes every 6 months. Suppose $S_0 = 100$, $u = 1.1$, $d = 0.95$, $X = 110$, $T = 1$ year, and $r_f = 5\%$. Find C.

We now have a two-stage stock tree:

After 6 months:

- Up: $uS_0 = 110$.
- Down: $dS_0 = 95$.

After a second 6 months:

- Up-Up: $u^2S_0 = 121$.
- Up-Down: $udS_0 = 95 \times 1.1 = 104.5$.
- Down-Up: $duS_0 = 95 \times 1.1 = 104.5$.
- Down-Down: $d^2S_0 = 90.25$.

The option values at $T = 1$ year are:

- $C_{uu} = \max(0, 121 - 110) = 11$.
- $C_{ud} = C_{du} = \max(0, 104.5 - 110) = 0$.
- $C_{dd} = \max(0, 90.25 - 110) = 0$.

Then, convert the two-stage problem into a series of one-stage problems.

- Treat $uS_0 = 110$ as the initial stock price and find the option price at $t = 0.5$ years.
- Treat $dS_0 = 95$ as the initial stock price and find the option price at $t = 0.5$ years.

Solve for C_u and C_d using any of the three methods.

Lecture 22, 2025/11/26

Review the last lecture.

Black-Scholes Model

Stocks have a constant volatility σ. Stocks are log normally distributed. Also, the EMH is true.

The Black-Scholes formula is

$$C = S_0\Phi(d_1) - Xe^{-rT}\Phi(d_2)$$

where

- X = strike price.

- T = time to expiry.
- r = risk-free rate (compounded continuously, i.e. c.c).
- $d_1 = \frac{\ln(S_0/X)+(r_f+\frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$.
- $d_2 = d_1 - \sigma\sqrt{T}$.
- $\Phi(\cdot)$ = CDF of standard normal distribution.

Example. Suppose $X = 150$, $S_0 = 160$, $r = 5\%$ c.c, $\sigma = 0.3$, and $T = 1$ year. Find C.

Proof. Calculate d_1 and d_2:

$$d_1 = 0.5282, \quad d_2 = 0.31602.$$

We are also given that the CDF values to get $C = 20.92$. □

7.3 Exotic Options

Here are some exotic options:

- **Bermudan Option**: can only be exercises on certain dates.
- **Look-back Option**: the payoff depends on the maximum stock price.
- **Asian Option**: the payoff depends on the average stock price.

Example.
Missing example.
As variability increases, the option price increases. What happens to C when r_f increases?

7.4 Forwards

Definition 7.5 (Forward Contract).
A **forward contract** gives us the right (and the obligation) to buy/sell a certain amount of goods at a pre-specified price at a future date.

- Buying the forward contract: long position.
- Selling the forward contract: short position.

Remark (**Fact**).
At the time of signing the forward contract, NPV = 0 because no money changes hands.

Example. Gold. Let S_0 = today's price and r_f = risk-free rate. Note that there is zero convenience yield and no storage cost.
Strategy 1: Borrow S_0 dollars and use that to buy gold. Now, the net cash flow at $t = 0$ is 0. On expiry, we have gold and need to pay back $S_0(1 + r_f)^T$ dollars.
Strategy 2: Buy the forward. The net cash flow at $t = 0$ is 0. On expiry, we need to pay $F(0, T)$ dollars to get gold.
By the law of one price, we have

$$F(0, T) = S_0(1 + r_f)^T.$$

Example (**Convenience Yield**).
Stocks. Let S_0 = today's price, D = PV of all dividends between $[0, T]$.
Strategy 1: Borrow S_0 dollars and use that to buy stocks. Now, the net cash flow at $t = 0$ is 0. On expiry at $t = T$, we have stocks and collect $D(1 + r_f)^T$ dollars from dividends. We need to pay back $S_0(1 + r_f)^T$ dollars, and pay back $S_0(1 + r_f)^T$ dollars.
Strategy 2: Long forward. On expiry at $t = T$, we need to pay $F(0, T)$ dollars to get stocks.
By the law of one price, we have

$$F(0, T) = S_0(1 + r_f)^T - D(1 + r_f)^T = (S_0 - D)(1 + r_f)^T.$$

Note. Convenience yield is one reason why a forward price can be lower than the spot price compounded at the risk-free rate.

Example. Storage costs. Let S_0 = today's price, and let C = PV of all storage costs between $[0, T]$.
Strategy 1: Borrow S_0 dollars and use that to buy wheat. On expiry at $t = T$, we have wheat

and need to pay back $S_0(1+r_f)^T$ dollars, and pay $C(1+r_f)^T$ dollars for storage costs.

Strategy 2: Long forward. On expiry at $t = T$, we need to pay $F(0, T)$ dollars to get wheat.

By the law of one price, we have

$$F(0,T) = S_0(1+r_f)^T + C(1+r_f)^T = (S_0 + C)(1+r_f)^T.$$

END OF ACTSC 372!

A Tutorials

A.1 Tutorial 1 (2025/09/19)

Mobius 2 Hints

Q1: Ignore CCA = cannot subtract depreciation.

Q10: $\sigma(NPV)$ is the standard deviation of NPV.

Note. For $Y = a + bX$, $\sigma_Y = |b|\sigma_X$.

Expected Utility

$U(X)$ = utility you get from lottery X. Under certain assumptions,

$$U(X) = \mathbb{E}[u(X)] = \sum u(x)f(x)$$

where $u(x)$ is the von-Neumann Morgenstern utility function and $f(x)$ is the PMF of X.

Risk Aversion

- A person is risk averse if $u(\mathbb{E}[X]) > \mathbb{E}[u(X)]$ for all X.
- A person is risk neutral if $u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$ for all X.
- A person is risk loving if $u(\mathbb{E}[X]) < \mathbb{E}[u(X)]$ for all X.

Note. $u(\cdot)$ is a concave function $\iff$ risk averse.

Example. Let X be a lottery with PMF $\mathbb{P}(X = 64) = \mathbb{P}(X = 8) = 0.5$ and let Y be a lottery with PMF $\mathbb{P}(Y = 125) = \mathbb{P}(Y = 0) = 0.5$. Consider $u(x) = x^{1/3}$ and $u(x) = \sqrt{x}$.
Note that for X with $u(x) = x^{1/3}$,

$$U(X) = 0.5 \cdot 64^{1/3} + 0.5 \cdot 8^{1/3} = 3.$$

For Y with $u(x) = x^{1/3}$,

$$U(Y) = 0.5 \cdot 125^{1/3} = 2.5.$$

Find $U(X)$ and $U(Y)$ for $u(x) = \sqrt{x}$.

Certainty Equivalent

Let $CE(X)$ = amount of dollars for sure that makes you indifferent with the lottery X.

- Risk averse: $CE(X) < \mathbb{E}[X]$.
- Risk neutral: $CE(X) = \mathbb{E}[X]$.
- Risk loving: $CE(X) > \mathbb{E}[X]$.

Definition A.1 (Certainty Equivalent).
The **certainty equivalent** of a lottery X is the amount of money $CE(X)$ such that

$$u(CE(X)) = \mathbb{E}[u(X)] .$$

If $u(\cdot)$ is strictly increasing, $CE(X) = u^{-1}(\mathbb{E}[u(X)])$.

Example. Consider the following:

- Wealth: \$100.
- Fine: \$36 with probability $p = \frac{1}{4}$.
- M: \$9.25, parking fee to avoid the fine.
- Utility function: $u(x) = \sqrt{x}$.

Proof. Pay for parking $= \sqrt{100 - M}$. Don't pay for parking $= \frac{1}{4}\sqrt{100-36} + \frac{3}{4}\sqrt{100} = 9.5$. So $\sqrt{100 - M} > 9.5 \implies M < 100 - 9.5^2 = 9.75$. So we should pay for parking. □

Are We Rational?

Example (Allais Paradox).

Consider the following choices:

- A: \$1 million dollars with $p = 1$.
- B: \$1 million dollars with $p = 0.89$, \$5 million dollars with $p = 0.1$, and \$0 with $p = 0.01$.
- C: \$0 with $p = 0.9$, \$1 million dollars with $p = 0.1$.
- D: \$0 with $p = 0.89$, \$1 million dollars with $p = 0.11$.

If you prefer A over B and C over D, you are not rational.

A.2 Tutorial 2 (2025/10/03)

I did not go.

A.3 Tutorial 3 (2025/11/14)

Intro to Options

Definition A.2 (**Derivative**).
A **derivative** is an asset whose value depends on the price of some other asset.

Definition A.3 (**Call Option**).
A **call option** is an asset which gives the buyer the right, but not an obligation, to buy an asset at a pre-determined price on/before a pre-determined date.

Notations:

- Underlying asset: stock.
- S_0 = price of the stock today.
- T = expiration date.
- S_T = price of the asset on the expiration date.
- X = pre-agreed price (i.e. strike price).
- Buyer: holder (long position).
- Seller: writer (short position).
- Price of the option = premium.
- "On the day": European option.
- "On/Before expiration date": American option.

Note. We assume that all options are European unless otherwise stated.

Value at Expiry

Example. Suppose $S_0 = 50, X = 60$, and $T = 1$ year.

If $S_T = 55$, then the value of the call option is 0.

If $S_T = 62$, then the value of the call option is 2.

Let V_T be the value of the call option on the expiration date. Then,

$$V_T = \begin{cases} 0, & S_T < X \\ S_T - X, & S_T \geq X. \end{cases}$$

$V_T = \max(0, S_T - X)$.

Definition A.4 (**Put-Option**).

A **put-option** gives the buyer the right (but not an obligation) to sell an asset at a pre-determined price on/before a pre-determined date of expiry.

Incomplete, but the lectures cover the same material.